

GWAS, dose-0 consistency, growth rate and IC50 on the DH4148 reference

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GWAS, dose-0 consistency, growth rate and IC50 on the DH4148 reference

Generated 2026-10-09. Phenotypes: Chromium, Copper and Lead screens (Iron and Zinc are held out, D-56). Genotypes: popgen callset on the DH4148 assembly (GCA_058775505.1). Panel: 126 haploid *R. mucilaginosa* strains (D-49). All numbers come from `report/tables/*.csv`. Scripts: `scripts/01` to `scripts/12`.

Summary

1. **The 126-strain panel is 14 clonal lineages.** Strains that differ by at most 2,000 SNPs form 14 groups (sizes 42, 34, 16, 6, 6, 6, 6, 3, 2 and five single strains). Within a lineage the median difference is 557 SNPs. Between lineages it is at least 48,641 SNPs. The 5-SNP de-cloning of D-49 does not remove this structure. The effective sample size for a genome-wide scan is about 14, not 126.
2. **Genome-wide hits are lineage markers.** With the relatedness matrix only, 64 loci pass the Bonferroni threshold (unadjusted phenotypes) and 44 loci pass it (run-adjusted phenotypes). **All of them are SNPs that do not vary inside any lineage.** They mark which lineage a strain belongs to. They cannot be mapped to a gene.
3. **Within lineages, nothing passes the threshold.** With run adjustment and lineage covariates the scans find 0 Bonferroni loci in 20 traits. 17 SNPs reach the suggestive level, against about 94 expected by chance. The 12 suggestive loci are listed for reference (Table 13). They are hypotheses, not results.
4. **Lineage explains part of some traits.** Intraclass correlation by lineage: Cr relative area at the top dose 0.29, Cr area AUC 0.28, Cu a* AUC 0.26, dose-0 ln area 0.24, dose-0 a* 0.23, Cu relative area at the top dose 0.20. Pb traits and all growth-rate traits are at most 0.10. With 14 lineages these estimates are uncertain.
5. **Non-additive variance is not detectable.** The strains are haploid, so there is no dominance. For epistasis, the additive-by-additive variance component is not significant in any of 20 traits (smallest $p = 0.095$). A scan of about 65,000 SNP pairs per trait found no pair below $p = 1e-5$ (about 0.65 expected by chance).
6. **Dose-0 (YPD) controls are consistent for a*, less so for size and not for growth rate.** Strain explains 38% of dose-0 a* variance across the three screens, and the screen itself explains about 0%. For ln area, strain explains 18% and the screen 17%. For the maximum growth rate, strain explains 3%.
7. **Growth rate and IC50.** Relative growth rate falls with dose in Cr (0.41 of control at dose 1.2) and Pb (0.12 at dose 30). Cu barely changes it (0.93 at dose 30). IC50 of relative area could be fitted for almost all strains in Cr and Pb. In Cu 138 of 317 strains are right-censored, because relative area stays at or above 0.5 at the top dose.

8. **No recombination inside the large lineages.** In the three largest lineages (42, 34 and 16 strains) the fraction of SNP pairs showing all four gametes does not change with distance: it is the same for pairs less than 1 kb apart, 500 kb apart and on different contigs (0.0% in L01, 1.7% to 3.0% in L02, 2.9% to 5.0% in L03). Recombination would make close pairs compatible and unlinked pairs incompatible. The pattern fits strict clonality (section 7).
9. **The Cr tree signal is partly a run effect.** Pagel’s lambda of baseline a^* in pure haploid *R. mucilaginosa* is 0.91 ($p = 0.008$, 163 strains). After removing run effects it is 0.89 ($p = 0.12$). The lineages with the lowest Cr a^* (L05, L07) were assayed mostly in the two later runs, and the later runs have lower a^* (section 8).
10. **Within lineages there is not enough signal to link Cr traits to loci.** In per-lineage permutation scans, 1 of 18 Cr tests (3 lineages, 6 traits) has a family-wise p below 0.05, which is what chance gives (0.9 expected). No candidate survives correction for the 60 lineage-by-trait tests. Candidate SNP and gene panels with window plots are in section 9. The metal and stress genes in the panels are not enriched (permutation $p = 0.78$).

1. Data

- **Phenotypes** come from the DuckDB `heavy_metal_measurement` table with strain, species and ploidy from the `strain_info` view (D-58). a^* is `ColorLab_a*GeoMedian`. The largest object per well and image is used. The late window and the 2,000 px colony cutoff follow the a^* report (`analysis/carotenoid_stress_vs_baseline`).
- **Growth rate** of a well is the largest slope of $\ln$ area against time over windows of 5 consecutive images. A window must start at 300 px or more and span at least 4 h. A well needs at least 6 images.
- **Relative traits** compare a strain with itself at dose 0: area AUC (relative area against dose scaled 0 to 1), relative area at the top dose, relative growth-rate AUC, a^* change at the top dose and a^* AUC.
- **IC50** is fitted for relative area with a two-parameter log-logistic curve. It is reported only when the curve crosses 0.5 inside the tested dose range.
- **GWAS traits** are 20 per strain (Table 1). Each is rank-normalised before the scan. Copper IC50 is not used (46% of panel strains are censored).
- **Genotypes.** Biallelic SNPs from popgen’s `rmuc_core` file, kept at minor allele frequency 0.05 or more and at most 10% missing genotypes inside the panel: 244,956 SNPs. After collapsing identical genotype patterns there are 7,199 distinct patterns (the effective number of tests for the lineage-free scans). With lineage covariates there are 2,940.
- **Annotation.** `snpEff` effects come from the VCF. Gene IDs and product names come from the DH4148 GFF. No GO or KEGG annotation exists for DH4148 yet.

trait	count	mean	std	min	50%	max
d0_a	125.000	0.896	2.060	-5.236	1.321	6.291
d0_lnA	125.000	-0.004	0.109	-0.542	0.011	0.251
d0_rgr	125.000	0.000	0.002	-0.005	0.000	0.004
Cr_da_top	125.000	-12.880	5.731	-20.941	-14.248	0.299
Cr_a_auc	125.000	-1.123	1.567	-4.887	-1.263	1.930
Cr_relarea_top	125.000	0.152	0.416	0.038	0.084	3.575
Cr_relarea_auc	125.000	0.591	0.067	0.450	0.577	0.897
Cr_relrgr_auc	124.000	0.880	0.135	0.662	0.847	1.271
Cr_logIC50	123.000	-0.471	0.105	-0.791	-0.477	-0.071
Cu_da_top	125.000	-15.411	3.007	-21.478	-15.299	-3.941
Cu_a_auc	125.000	-6.333	2.197	-13.938	-6.017	1.586
Cu_relarea_top	125.000	0.547	0.236	0.105	0.489	1.240
Cu_relarea_auc	125.000	1.006	0.213	0.549	0.988	2.377
Cu_relrgr_auc	125.000	1.117	0.166	0.845	1.100	1.681
Pb_da_top	123.000	-11.111	7.199	-19.034	-13.645	10.347
Pb_a_auc	123.000	-0.822	3.121	-6.732	-1.587	7.057
Pb_relarea_top	123.000	0.128	0.109	0.046	0.085	0.585
Pb_relarea_auc	123.000	0.542	0.125	0.336	0.506	0.947
Pb_relrgr_auc	122.000	0.734	0.148	0.438	0.708	1.171
Pb_logIC50	119.000	2.580	0.114	2.102	2.591	2.819

Table 1. GWAS traits in the 126-strain panel (raw values; 119-125 strains per trait).

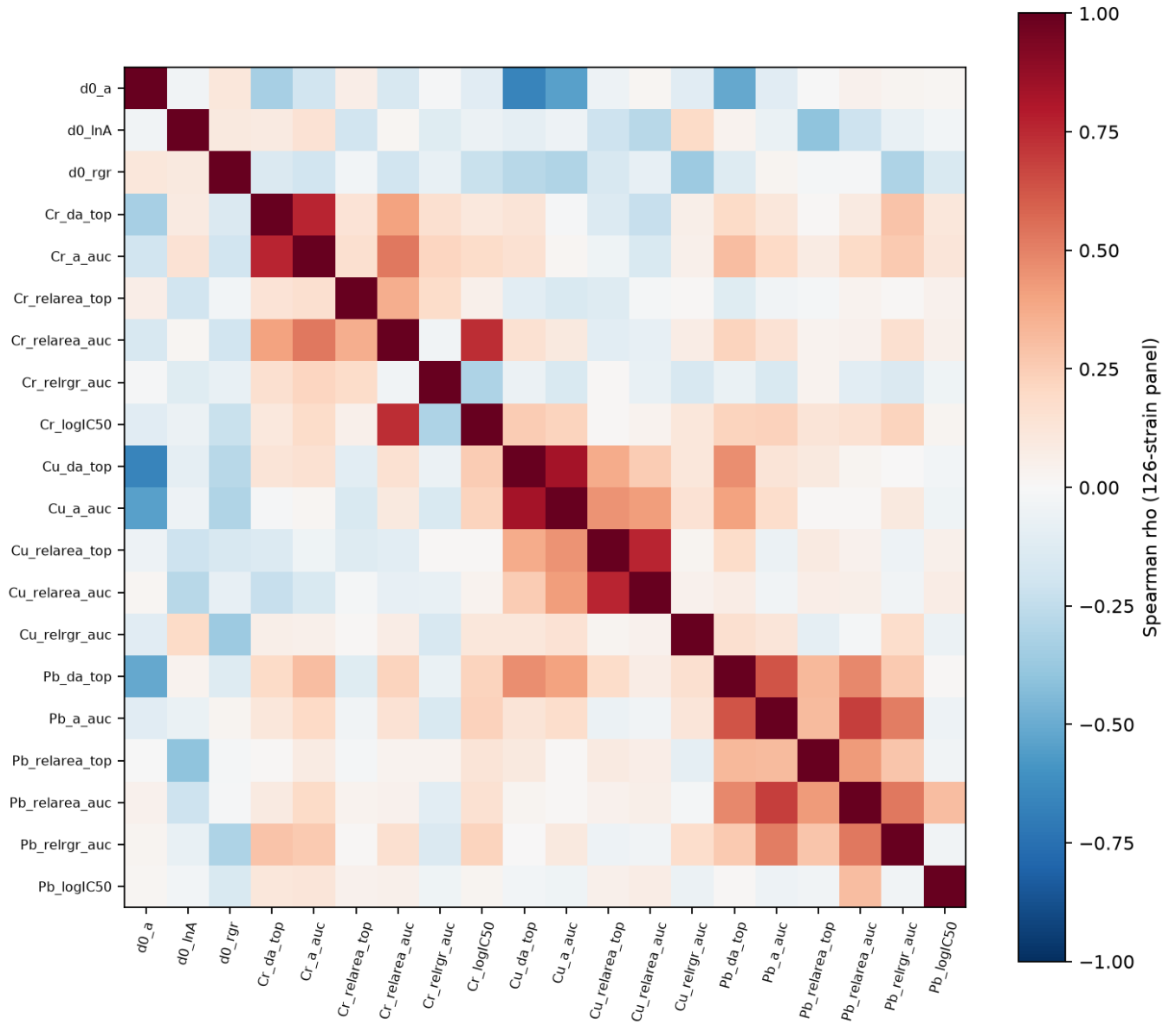


Figure 1. Spearman correlation among the 20 traits (panel strains).

2. Dose-0 (YPD) consistency across experiments

Dose-0 wells of the three screens are pooled. Model: $\text{trait} \sim 1 + (1 \mid \text{strain}) + (1 \mid \text{screen}) + (1 \mid \text{run}) + (1 \mid \text{plate})$. Wells are replicate wells on dose-0 plates. Fe and Zn dose-0 plates exist but are partial, so the main model uses Cr, Cu and Pb. A version with all five screens is in `report/tables/dose0_variance_components.csv`.

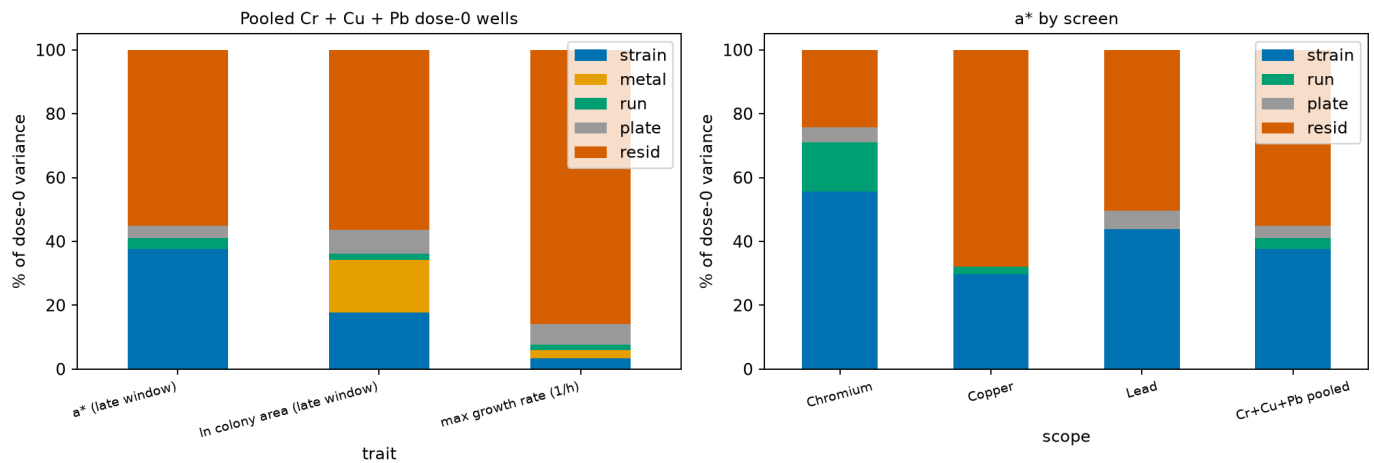


Figure 2. Shares of dose-0 variance: strain, screen (metal), run, plate and residual.

scope	trait	n_wells	n_strains	share_strain	share_metal	share_run	share_plate	share_resid
Cr+Cu+Pb pooled	a	3282	319	0.38	0.00	0.03	0.04	0.55
Chromium	a	1188	306	0.56		0.15	0.05	0.24
Copper	a	1116	319	0.30		0.02	0.00	0.68
Lead	a	978	297	0.44		0.00	0.06	0.50
Cr+Cu+Pb pooled, GWAS panel strains	a	1371	125	0.28	0.00	0.01	0.08	0.62
Cr+Cu+Pb pooled	lnA	3282	319	0.18	0.16	0.02	0.07	0.56
Chromium	lnA	1188	306	0.58		0.09	0.04	0.30
Copper	lnA	1116	319	0.13		0.02	0.06	0.80
Lead	lnA	978	297	0.29		0.00	0.13	0.58
Cr+Cu+Pb pooled, GWAS panel strains	lnA	1371	125	0.13	0.20	0.01	0.11	0.55
Cr+Cu+Pb pooled	rgr	3412	320	0.03	0.03	0.02	0.06	0.86
Chromium	rgr	1194	309	0.07		0.10	0.02	0.81
Copper	rgr	1141	319	0.04		0.00	0.06	0.90
Lead	rgr	1077	307	0.14		0.00	0.08	0.78
Cr+Cu+Pb pooled, GWAS panel strains	rgr	1405	125	0.02	0.04	0.03	0.06	0.85

Table 2. Variance shares of dose-0 wells. *share_metal* is empty for single-screen models. Fits marked singular in the table have a variance component at zero.

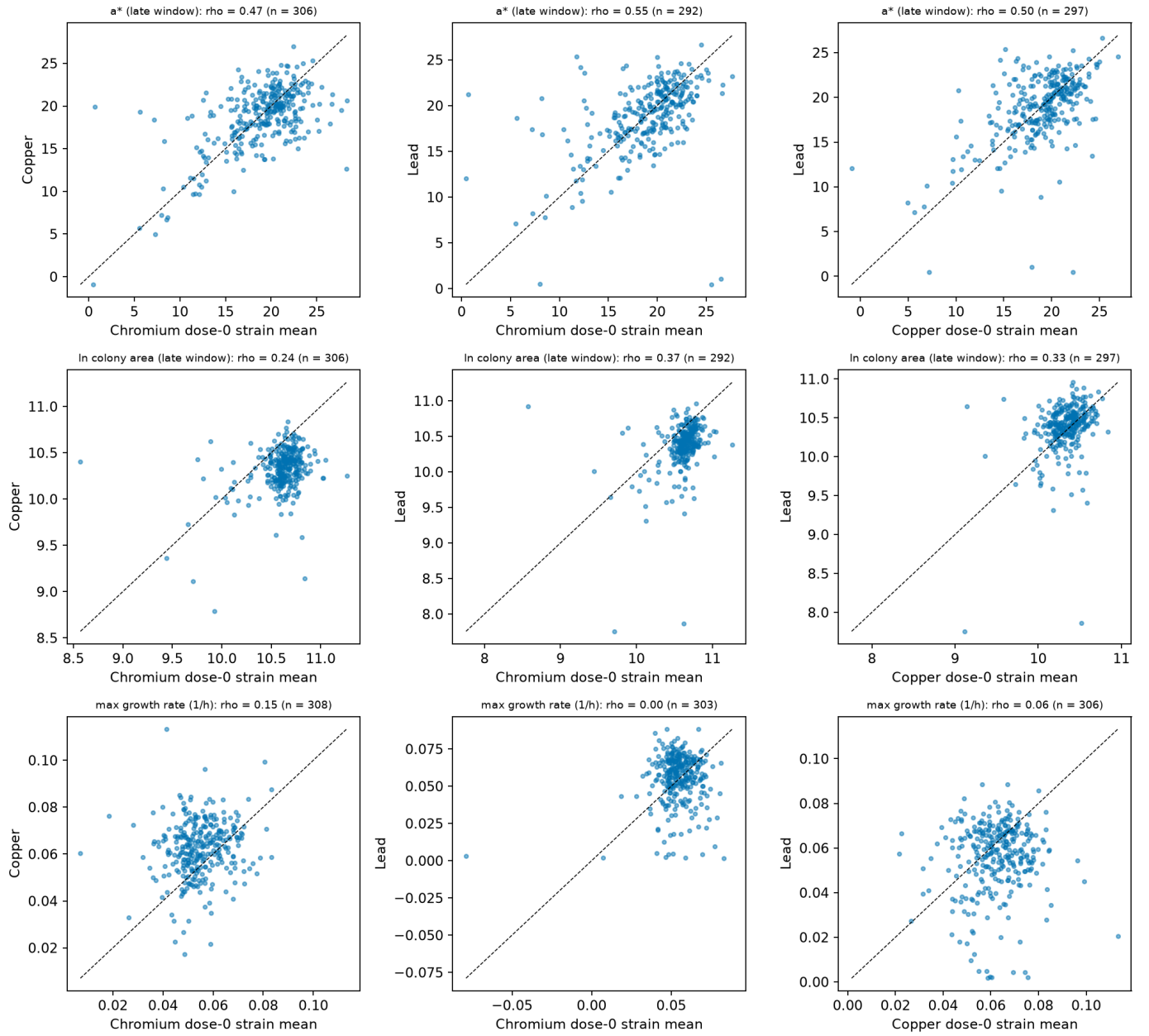


Figure 3. Dose-0 strain means in one screen against another (dashed = equal values).

trait	screen_1	screen_2	n_strains	spearman	pearson
a	Chromium	Copper	306	0.47	0.59
a	Chromium	Lead	292	0.55	0.48
a	Copper	Lead	297	0.50	0.55
lnA	Chromium	Copper	306	0.24	0.32
lnA	Chromium	Lead	292	0.37	0.32
lnA	Copper	Lead	297	0.33	0.33
rgr	Chromium	Copper	308	0.15	0.14
rgr	Chromium	Lead	303	0.00	0.11
rgr	Copper	Lead	306	0.06	0.06

Table 3. Agreement of dose-0 strain means between screens.

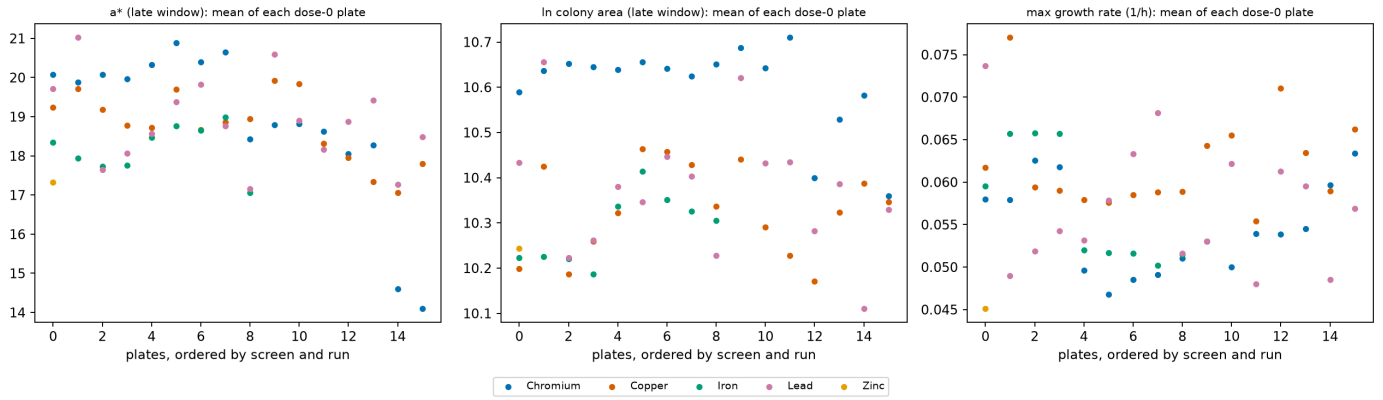


Figure 4. Mean of each dose-0 plate, by screen.

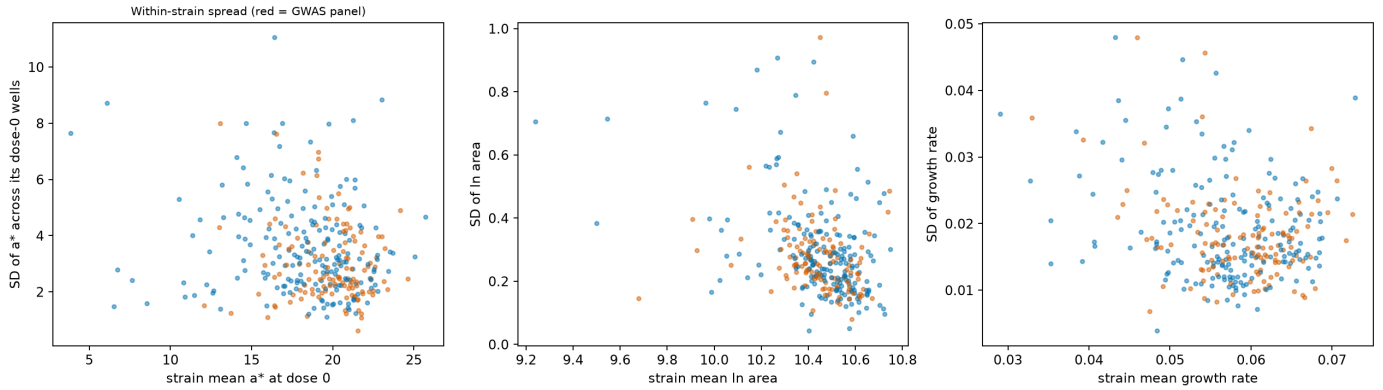


Figure 5. Spread of a^* , ln area and growth rate across the dose-0 wells of each strain (red = GWAS panel).*

sample_name	species	n_wells	a_mean	a_sd	a_sd_z
TFCN_25-333Y-2	R. sp_clade_I	11	16.43	11.05	5.88
TFCN_25-333D-1	R. sp_clade_I	8	23.04	8.83	4.26
DBVPG_6660	R. frigidialcoholis	10	6.11	8.71	4.18
TFCN_86A-8	R. sphaerocarpa	11	21.28	8.10	3.73
DBVPG_4952	R. mucilaginosa	7	16.91	7.99	3.65
DBVPG_6083	R. graminis	10	14.68	7.99	3.65
DBVPG_3236	R. mucilaginosa	11	13.08	7.99	3.65
TFCN_86A-5	R. sphaerocarpa	11	19.77	7.97	3.64
TFCN_86C-10	R. sphaerocarpa	7	16.40	7.66	3.41
NRRL_Y-7192	R. sphaerocarpa	11	3.87	7.64	3.40
DBVPG_4203	R. mucilaginosa	11	16.55	7.61	3.38
DBVPG_3769	R. dairenensis	12	18.64	7.33	3.17

Table 4. Strains whose dose-0 wells vary most in a^* (robust z of the SD across wells; strains with at least 6 wells).

Reading.

- **a^* at dose 0 is a reproducible strain trait.** Strain explains 38% of the variance in the pooled model, and the screen explains about 0%. Strain means agree between screens (Spearman 0.47 to 0.55). Run and plate explain 3% and 4%. Half of the variance (55%) is replicate-well noise.
- **Colony size at dose 0 differs between screens.** The screen explains 17% of ln area variance, against 18% for strain. Plate means of ln area are 10.60 (Cr), 10.33 (Cu) and 10.37 (Pb). Strain means agree only weakly between screens (0.24 to 0.38).
- **Growth rate is not a reproducible strain trait here.** Strain explains 3% of its variance and between-screen correlations are 0.00 to 0.15. It is not suitable for mapping.
- **Cr dose-0 plates vary most in a^* .** In the Cr-only model run explains 15% of the variance, against 2% in Cu and 0% in Pb. The plate-mean SD of a^* is 1.98 in Cr, 0.87 in Cu and 1.10 in Pb.

- **Within-strain spread.** The median SD of a^* across a strain's dose-0 wells is 2.98. 13 of 312 strains have a robust z above 3 (Table 4). Spread alone does not show mixed cultures. The identity columns of `strain_curation.csv` (`identity_flag_categories`) should be checked for these strains; this has not been done here.
- **Panel strains.** In the panel strain subset the strain share of a^* is 28% (all strains 38%).

3. Growth rate and IC50

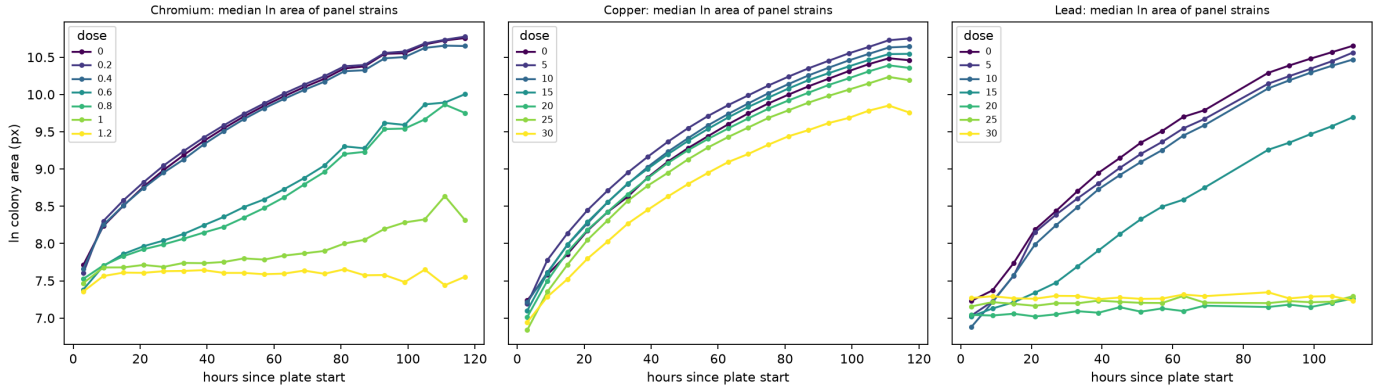


Figure 6. Median ln colony area of panel strains against time, by dose.

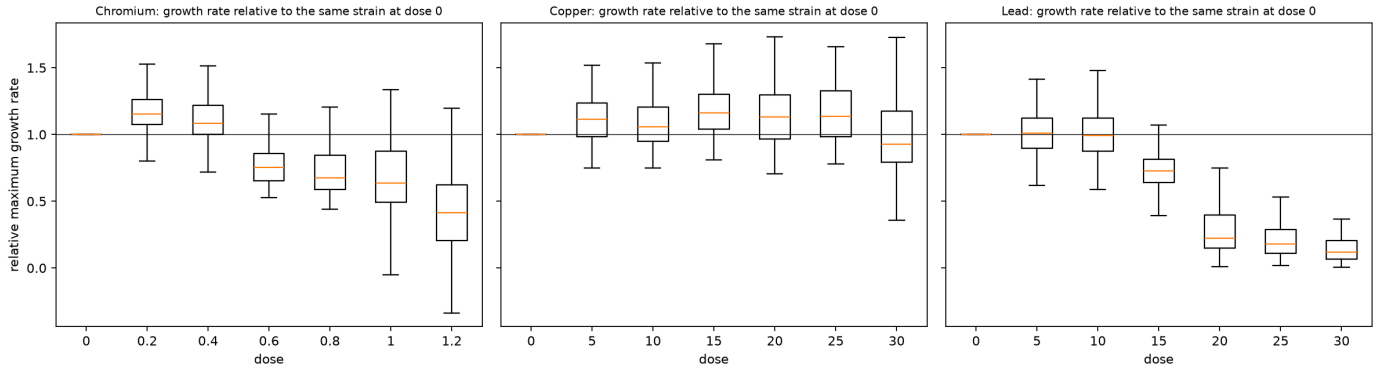


Figure 7. Maximum growth rate of each strain relative to the same strain at dose 0, by dose (panel strains).

Metal	dose	n_strains	median_rel_rgr	q25	q75
Chromium	0.00	125	1.00	1.00	1.00
Chromium	0.20	125	1.15	1.07	1.26
Chromium	0.40	125	1.08	1.00	1.22
Chromium	0.60	125	0.75	0.65	0.86
Chromium	0.80	125	0.68	0.59	0.84
Chromium	1.00	124	0.64	0.49	0.88
Chromium	1.20	102	0.41	0.20	0.62
Copper	0.00	125	1.00	1.00	1.00
Copper	5.00	125	1.11	0.98	1.23
Copper	10.00	125	1.06	0.95	1.20
Copper	15.00	125	1.16	1.04	1.30
Copper	20.00	125	1.13	0.97	1.30
Copper	25.00	125	1.14	0.98	1.33
Copper	30.00	125	0.93	0.79	1.17
Lead	0.00	123	1.00	1.00	1.00
Lead	5.00	123	1.01	0.90	1.12
Lead	10.00	123	0.99	0.88	1.12
Lead	15.00	123	0.73	0.64	0.81
Lead	20.00	67	0.22	0.15	0.40
Lead	25.00	76	0.18	0.11	0.29

Metal	dose	n_strains	median_rel_rgr	q25	q75
Lead	30.00	81	0.12	0.07	0.20

Table 5. Relative maximum growth rate by dose (median and quartiles across strains). Pb at doses 20 to 30 has only 67 to 81 strains with enough images for a growth rate.

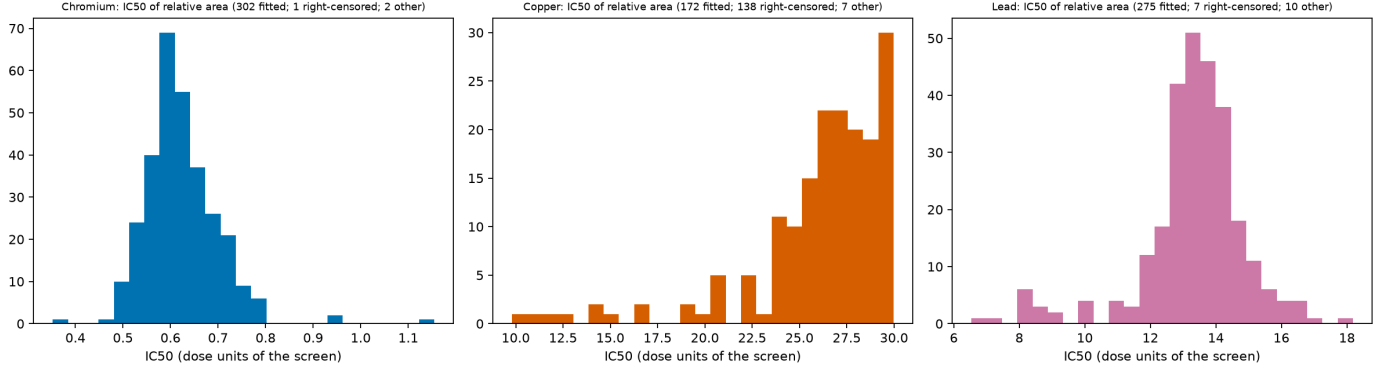


Figure 8. IC50 of relative area (all strains).

Metal	ok	outside_range	right_censored
Chromium	302	2	1
Copper	172	7	138
Lead	275	10	7

Table 6. IC50 fit status by metal (all strains).

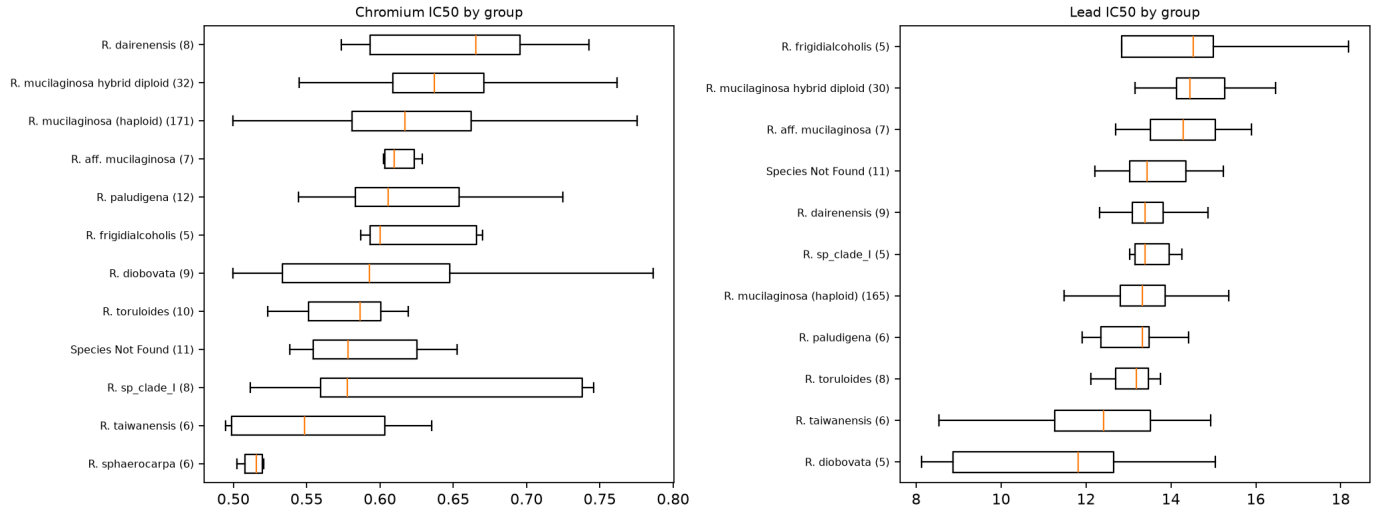


Figure 9. IC50 by species and ploidy group (groups with at least 5 fitted strains; Cr and Pb).

Metal	group	n	median_ic50	q25	q75
Chromium	R. aff. mucilaginosa	7	0.61	0.60	0.62
Chromium	R. dairenensis	8	0.67	0.59	0.70
Chromium	R. diobovata	9	0.59	0.53	0.65
Chromium	R. frigidialcoholis	5	0.60	0.59	0.67
Chromium	R. mucilaginosa (haploid)	171	0.62	0.58	0.66
Chromium	R. mucilaginosa hybrid diploid	32	0.64	0.61	0.67
Chromium	R. paludigena	12	0.61	0.58	0.65

Metal	group	n	median_ic50	q25	q75
Chromium	R. sp_clade_I	8	0.58	0.56	0.74
Chromium	R. sphaerocarpa	6	0.52	0.51	0.52
Chromium	R. taiwanensis	6	0.55	0.50	0.60
Chromium	R. toruloides	10	0.59	0.55	0.60
Chromium	Species Not Found	11	0.58	0.55	0.63
Copper	R. aff. mucilaginosa	5	27.16	25.58	29.59
Copper	R. dairenensis	9	27.92	22.19	28.50
Copper	R. diobovata	6	26.66	24.28	28.28
Copper	R. mucilaginosa (haploid)	86	27.49	25.28	29.10
Copper	R. mucilaginosa hybrid diploid	20	26.97	26.01	27.54
Copper	R. paludigena	11	25.03	23.90	28.37
Copper	R. sp_clade_I	5	27.80	25.53	28.23
Copper	R. toruloides	7	26.54	26.06	28.59
Lead	R. aff. mucilaginosa	7	14.28	13.52	15.04
Lead	R. dairenensis	9	13.39	13.10	13.82
Lead	R. diobovata	5	11.82	8.86	12.65
Lead	R. frigidialcoholis	5	14.53	12.84	15.00
Lead	R. mucilaginosa (haploid)	165	13.33	12.80	13.86
Lead	R. mucilaginosa hybrid diploid	30	14.44	14.13	15.27
Lead	R. paludigena	6	13.32	12.35	13.48
Lead	R. sp_clade_I	5	13.39	13.15	13.96
Lead	R. taiwanensis	6	12.41	11.26	13.52
Lead	R. toruloides	8	13.18	12.70	13.47
Lead	Species Not Found	11	13.43	13.03	14.36

Table 7. IC50 (dose units of the screen) by group. Copper IC50 values are close to the top dose (30) and half of the strains are censored, so Copper IC50 is coarse.

Reading.

- **Cr:** growth rate rises at low doses (1.15 at 0.2) and falls to 0.64 at dose 1.0 and 0.41 at dose 1.2. Median IC50 is about 0.6 dose units for pure haploid *R. mucilaginosa* (IQR 0.58 to 0.66).
- **Cu:** growth rate stays at 0.93 to 1.16 of control through dose 30. Relative area falls only at the top doses. Most Cu IC50 values are outside the tested range.
- **Pb:** growth rate is unchanged to dose 10, 0.73 at dose 15, and 0.12 at dose 30. Median IC50 is about 13.3 for pure haploid strains.
- **Group differences.** *R. sphaerocarpa* and *R. taiwanensis* have the lowest Cr IC50 (0.52 and 0.55, 6 strains each). Hybrid diploids have a higher Pb IC50 (14.4) than pure haploids (13.3). Group sizes are small (5 to 12 strains), so these are descriptive.
- **IC50 against other traits.** Cr IC50 correlates with Cr area AUC (rho 0.74) but not with the a* change (0.10). Pb IC50 correlates weakly with Pb area AUC (0.31).

4. Genetic structure: 14 lineages

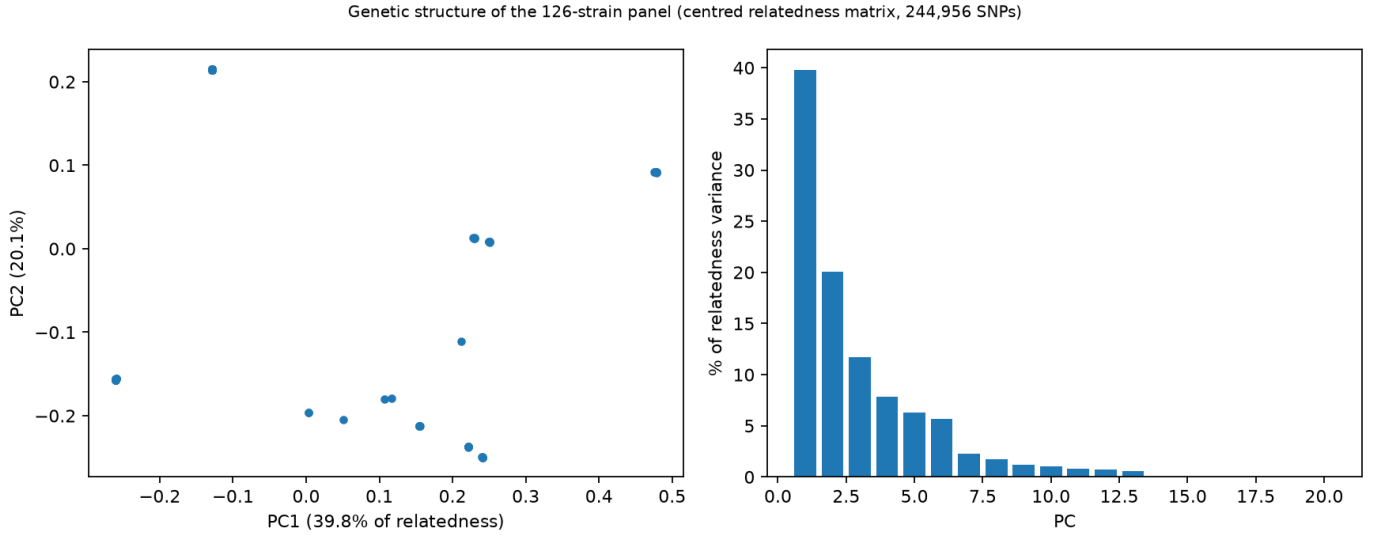


Figure 10. First two principal components of the centred relatedness matrix and the variance of the first 20 components. Each visible point is a cluster of near-identical strains.

max_snp_differences_within_group	n_groups	largest_groups
5	126	1, 1, 1, 1, 1, 1
50	93	10, 7, 5, 4, 4, 3
500	35	41, 14, 8, 6, 6, 6
2000	14	42, 34, 16, 6, 6, 6
5000	14	42, 34, 16, 6, 6, 6
20000	14	42, 34, 16, 6, 6, 6
50000	13	76, 16, 6, 6, 6, 6

Table 8a. Number of single-linkage groups of panel strains by the largest allowed SNP difference. The count is 14 for any cut from 2,000 to 20,000 differences.

lineage	n_strains	example_strains
L01	42	TFCN_98A-5, TFCN_102C-2, TFCN_105A-2, TFCN_17-334Y-2
L02	34	TFCN_107A-6, TFCN_102D-1, TFCN_4M-1-2, TFCN_2M-1-4
L03	16	DBVPG_3775, DBVPG_3776, DBVPG_3777, DBVPG_4920
L04	6	DBVPG_3381, DBVPG_3382, DBVPG_3383, DBVPG_3384
L05	6	DBVPG_3983, DBVPG_3984, DBVPG_3044, DBVPG_3090
L06	6	DBVPG_5228, DBVPG_5229, DBVPG_5235, DBVPG_10882
L07	6	DBVPG_4408, DBVPG_4376, DBVPG_5757, DBVPG_5758
L08	3	DBVPG_4945, DBVPG_4947, DBVPG_10842
L09	2	DBVPG_4958, DBVPG_10937
L10	1	TFCN_17-329Y-4
L11	1	DBVPG_4378
L12	1	DBVPG_4618
L13	1	TFCN_25-334D-3
L14	1	DBVPG_4377

Table 8b. Lineage sizes (lineage 1 is the largest).

- The first two components explain 40% and 20% of the relatedness. The panel has visible clusters, not a continuum.
- Lineage and run are associated in all three screens (chi-square $p < 0.001$). Strains of one lineage were often on the same runs. Removing run effects therefore removes part of the lineage signal as well.

Lineage as a source of trait variance

trait	ICC raw	ICC run-adjusted	p (run-adjusted, ANOVA)
Cr_a_auc	0.32	0.14	0.01
Cr_da_top	0.14	0.07	0.11
Cr_logIC50	0.14	0.12	0.02
Cr_relarea_auc	0.27	0.28	0.00
Cr_relarea_top	0.32	0.29	0.00
Cr_relrgr_auc	0.11	0.07	0.12
Cu_a_auc	0.25	0.26	0.00
Cu_da_top	0.17	0.17	0.00
Cu_relarea_auc	0.02	0.02	0.33
Cu_relarea_top	0.18	0.20	0.00
Cu_relrgr_auc	-0.06	-0.05	0.82
Pb_a_auc	0.04	0.05	0.16
Pb_da_top	0.03	0.07	0.10
Pb_logIC50	0.02	0.04	0.20
Pb_relarea_auc	-0.02	0.00	0.44
Pb_relarea_top	-0.00	0.08	0.09
Pb_relrgr_auc	-0.03	-0.05	0.83
d0_a	0.23	0.23	0.00
d0_lnA	0.24	0.24	0.00
d0_rgr	0.10	0.10	0.05

Table 9. Intraclass correlation of strain-level traits by lineage (one-way ANOVA estimator, 14 lineages), before and after removing run effects (fixed run effects, lineage as a random group).

- Lineage explains up to 29% of a trait (Cr relative area). With 14 lineages each ICC has a wide interval. This is a broad-sense measure: it mixes additive and non-additive genetic effects and anything else that differs between lineages.
- Pb traits and the growth-rate traits show no lineage component (ICC at most 0.10).

5. GWAS

Three scans per trait (GEMMA, univariate LMM, Wald test, centred relatedness matrix from all 244,956 SNPs):

- (a) relatedness only, unadjusted phenotypes;
- (b) relatedness, phenotypes with run effects removed;
- (c) as (b) plus lineage indicator covariates, so that only variation inside lineages is tested.

Bonferroni thresholds use the number of distinct genotype patterns (7,199 for (a) and (b), $p < 6.95e-6$; 2,940 for (c), $p < 1.7e-5$). The suggestive level is 1 / that number.

trait	OLS, no relatedness	LMM (K only)	LMM, run-adjusted	LMM, run-adjusted + lineage	Bonferroni SNPs: LMM (K only)	Bonferroni SNPs: LMM, run-adjusted	Bonferroni SNPs: LMM, run-adjusted + lineage
d0_a	3.18	0.95	0.95	1.02	0	0	0
d0_lnA	1.10	1.50	1.50	1.39	0	0	0
d0_rgr	3.94	1.63	1.63	0.95	0	0	0
Cr_da_top	2.03	1.07	0.82	1.33	0	0	0
Cr_a_auc	3.31	1.24	1.09	0.74	0	0	0
Cr_relarea_top	5.37	1.03	1.29	1.04	31	0	0
Cr_relarea_auc	3.81	1.34	1.03	1.27	845	0	0
Cr_relrgr_auc	0.99	1.08	0.88	1.14	0	0	0
Cr_logIC50	3.72	1.50	1.42	1.21	0	0	0
Cu_da_top	3.21	1.06	1.04	0.31	0	0	0
Cu_a_auc	2.82	1.09	1.19	0.52	0	0	0
Cu_relarea_top	4.77	0.72	0.73	0.54	8009	10099	0

trait	OLS, no related- ness	LMM (K only)	LMM, run- adjusted	LMM, run-adjusted + lineage	Bonferroni SNPs: LMM (K only)	Bonferroni SNPs: LMM, run-adjusted	Bonferroni SNPs: LMM, run-adjusted + lineage
Cu_relaea_top	2.78	1.10	1.27	0.75	0	0	0
Cu_relgr_top	1.75	1	1.49	1.15	0	0	0
Pb_da_top	2.01	1.30	1.23	0.87	0	0	0
Pb_a_auc	3.80	1.06	1.10	1.12	0	0	0
Pb_relaea_top	2.15	0.58	1.06	1.31	0	0	0
Pb_relaea_auc	0.70	2.50	0.58	0.81	0	0	0
Pb_relgr_top	0.44	0.75	0.34	1.27	0	0	0
Pb_logIC50	1.096	1.57	1.32	0.98	0	0	0

Table 10. Genomic inflation (λ) of the scans, and the number of SNPs below the Bonferroni threshold. “OLS, no relatedness” is the inflation of a plain regression on the run-adjusted phenotypes.

scan	loci_suggestive	loci_bonferroni	bonferroni_lineage_markers	contigs_with_bonferroni	loci_lineages_carrying_alt
unadjusted	174	64	64	15	8.00
runadj	165	44	44	9	7.00
runadj_lineage	12	0	0	0	

Table 11. Lead SNPs of the loci (SNPs within 50 kb are one locus) that are lineage markers: SNPs without variation inside any lineage.

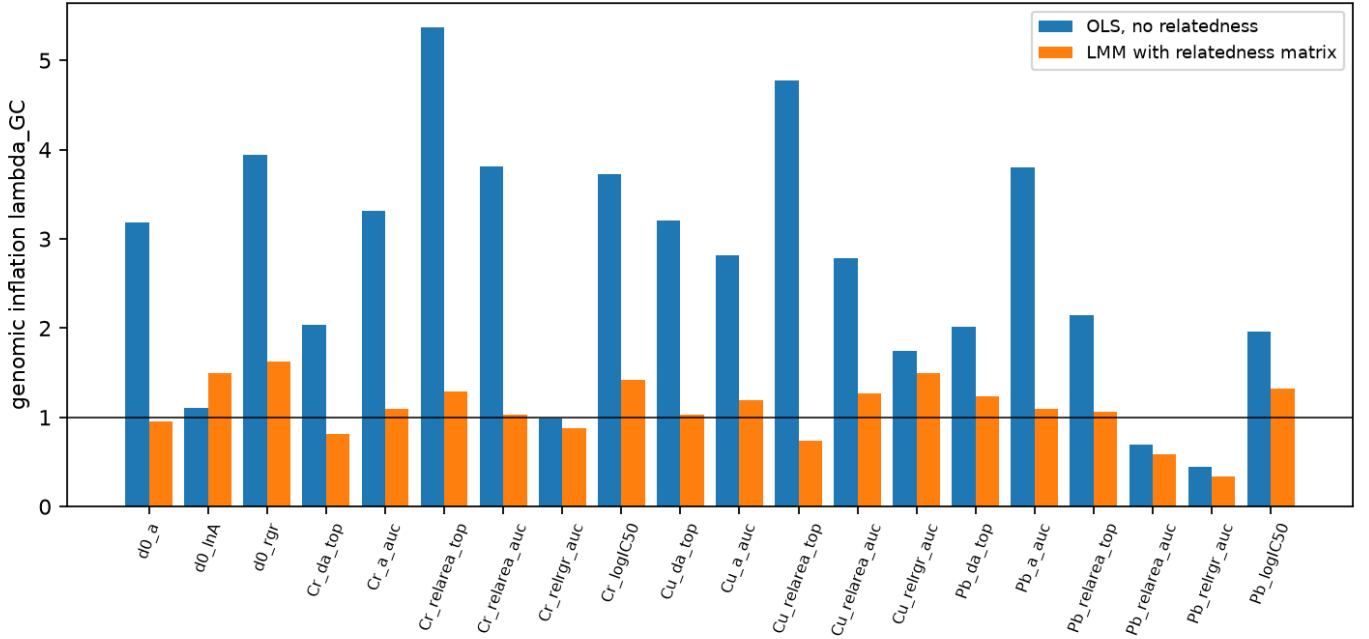


Figure 11. Inflation with and without the relatedness matrix (run-adjusted phenotypes).

Relatedness only (scans a and b)

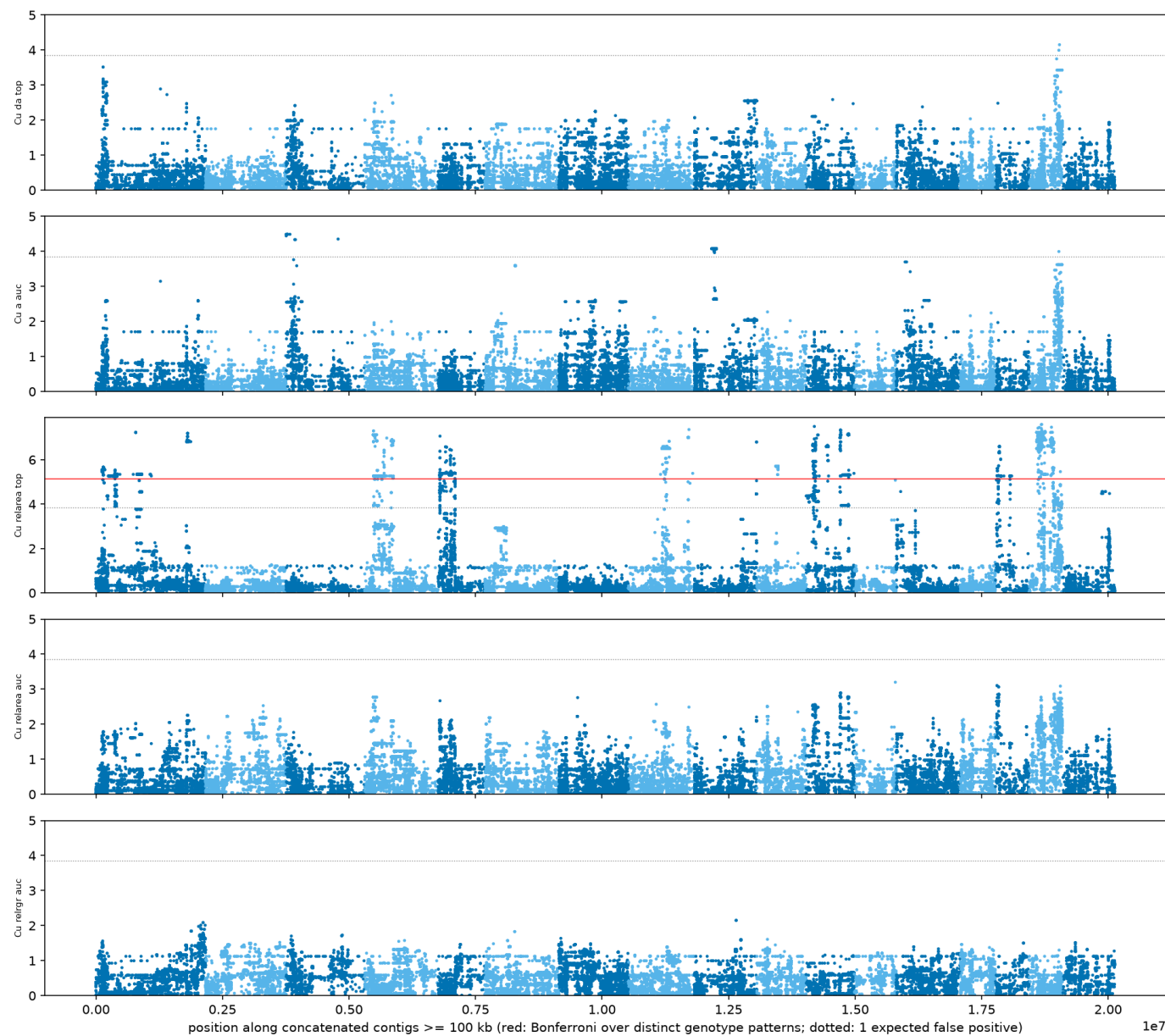


Figure 12. *Cu* traits, run-adjusted, relatedness only. The wide vertical blocks are sets of perfectly linked SNPs that mark lineages.

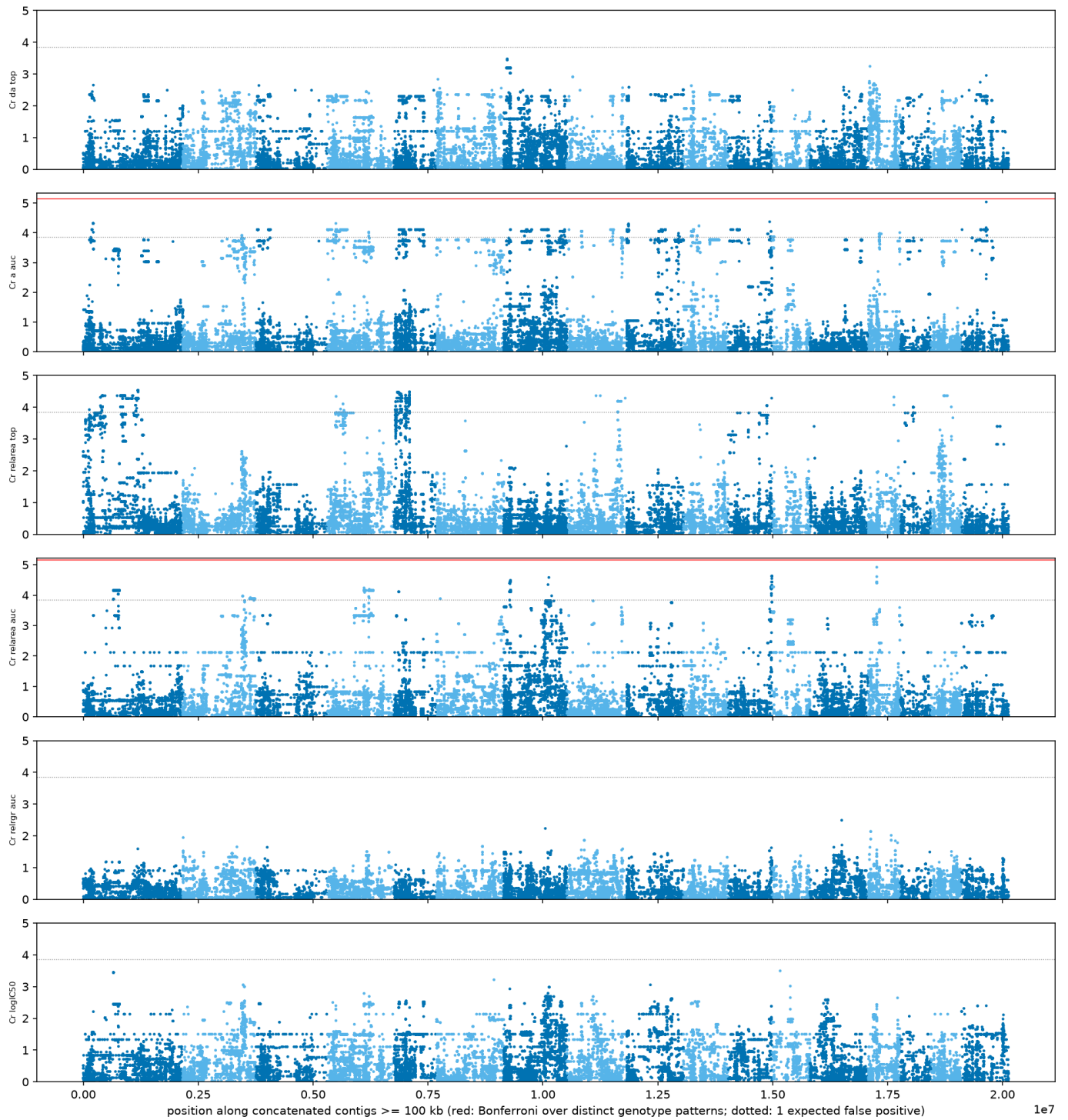


Figure 13. *Cr* traits, run-adjusted, relatedness only.

trait	chr	pos	af	beta	p_wald	suggestive	snps	in	regional	carrying	lineage	level	marked	product
Cu_relarea	CM17949748	10820.192-		4.0e- 938		7				True			ACY3AU_N02001	reductase
Cu_relarea	CM17948873	3030.225-		5.9e- 379		10				True			ACY3AU_3000108	reductase
Cu_relarea	CM17949807	7770.192-		6.1e- 205		5				True			ACY3AU_hyp0528	chemical protein
Cu_relarea	CM17948756	6040.198-		6.8e- 448		6				True			ACY3AU_hyp0528	chemical protein

trait	chr	pos	af	beta	p_wald	h_suggestive_snps	lineages_carrying	lineage	gene	product
Cu_relarea	CMd179491196734.208	-	7.5e- 61	0.926 08		9	True	ACY3AU_005187	met-tetrahydrofolate reductase (NAD	
Cu_relarea	CMd17949369830.199	-	7.9e- 348	0.938 08		6	True	ACY3AU_006015	metabolic protein	
Cu_relarea	CMd17949870850.195	-	8.3e- 983	0.94 08		7	True	ACY3AU_002885	PUF family protein 1	
Cu_relarea	CMd17949645600.2	-	8.4e- 1215	0.928 08		7	True	ACY3AU_002076	metabolic protein	
Cu_relarea	CMd17949794010.202	-	8.7e- 420	0.93 08		8	True	ACY3AU_002021	metabolic protein	
Cu_relarea	CMd17949860080.204	-	8.9e- 198	0.924 08		8	True	ACY3AU_004301	metabolic protein	
Cu_relarea	CMd17949366940.196	-	9.3e- 600	0.929 08		6	True	ACY3AU_005308	met-tRNA ligase	
Cu_relarea	CMd1794988399	0.198	9.7e- 447	0.961 08		7	True	ACY3AU_000013	metabolic protein	

Table 12. The 12 strongest Bonferroni loci of scan (a). Every one is a lineage marker: it carries the alternate allele in a set of whole lineages. The gene is the one at or nearest the lead SNP, and the gene is not the cause.

- A locus here is a statement about which lineages differ in the trait, and not about a gene. All SNPs on a lineage’s backbone have the same p-value. The lead SNP sits at an arbitrary position along the genome.
- The relatedness matrix does not solve this. With 14 lineages, a lineage-level trait difference is perfectly explained by thousands of SNPs.

Within lineages (scan c)

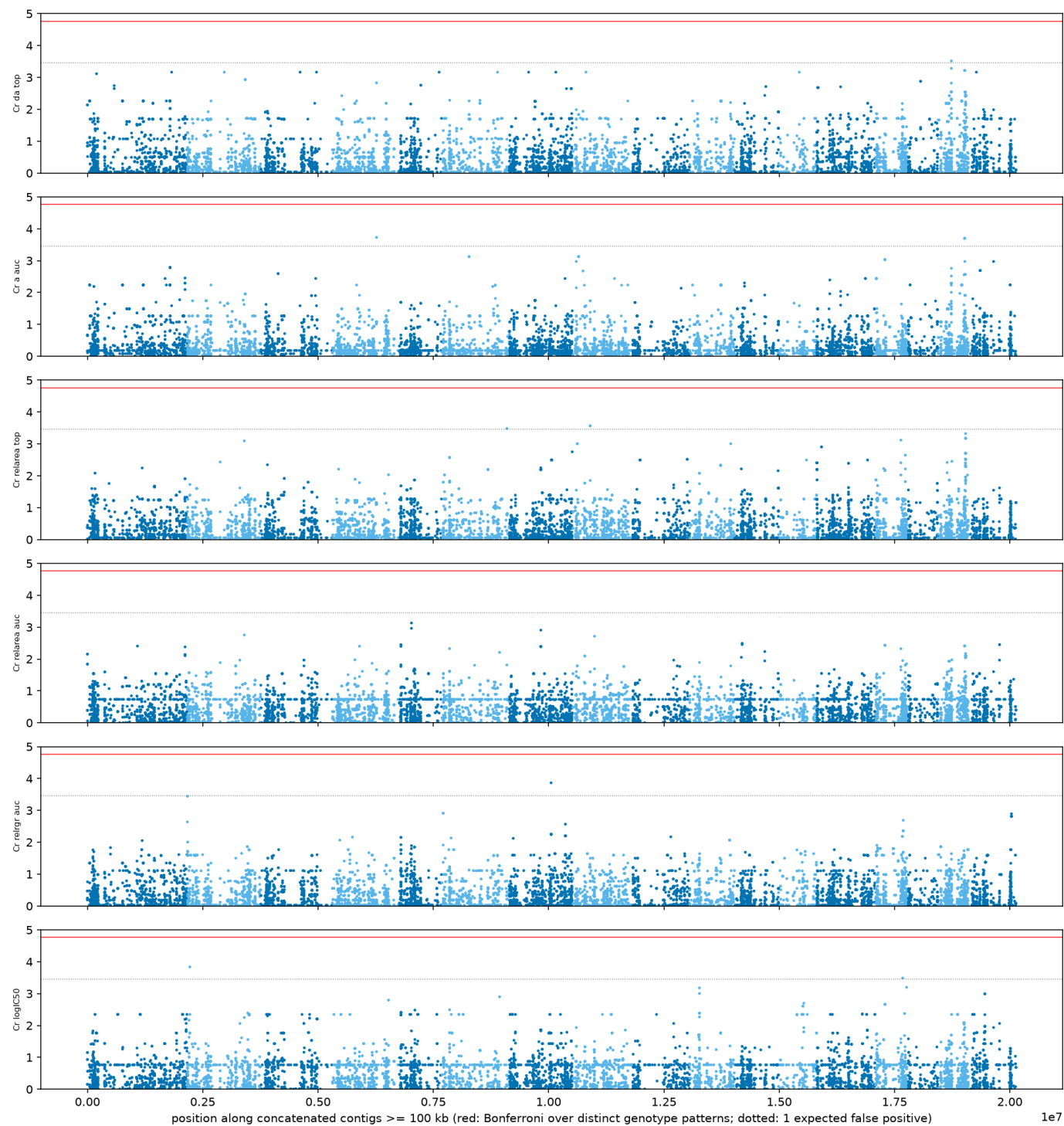


Figure 14. *Cr* traits, run-adjusted, lineage covariates.

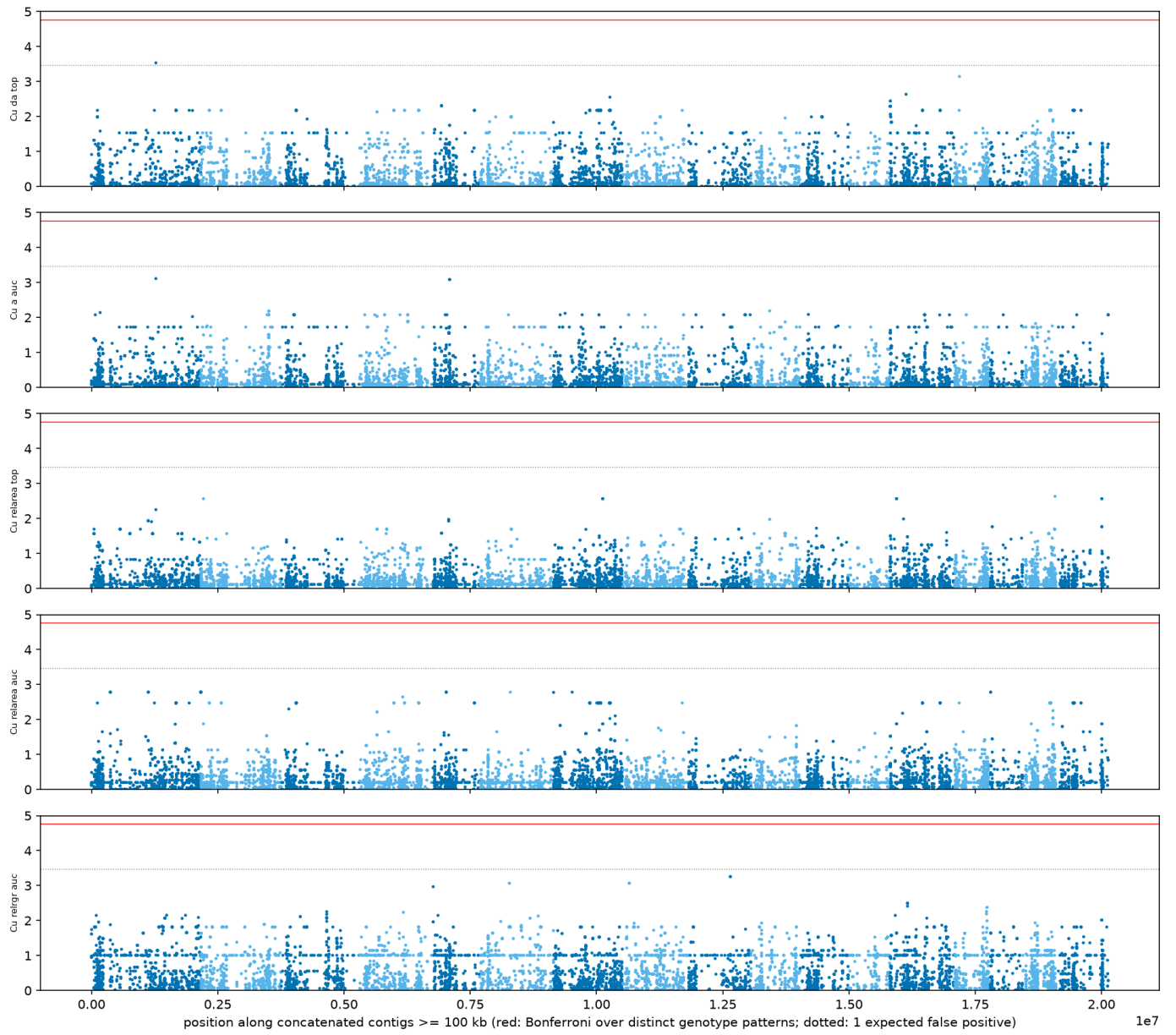


Figure 15. *Cu* traits, run-adjusted, lineage covariates.

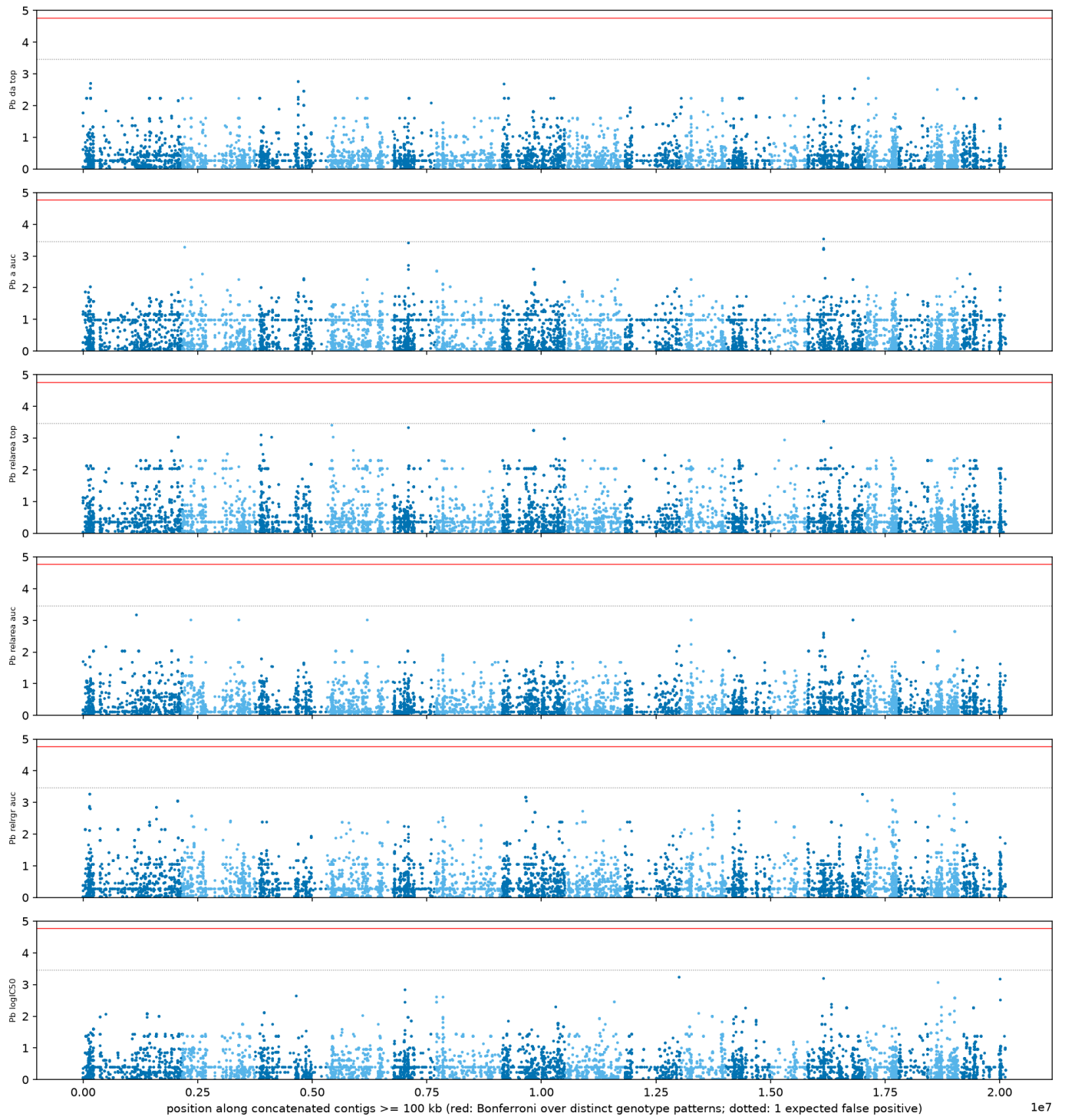


Figure 16. *Pb* traits, run-adjusted, lineage covariates.

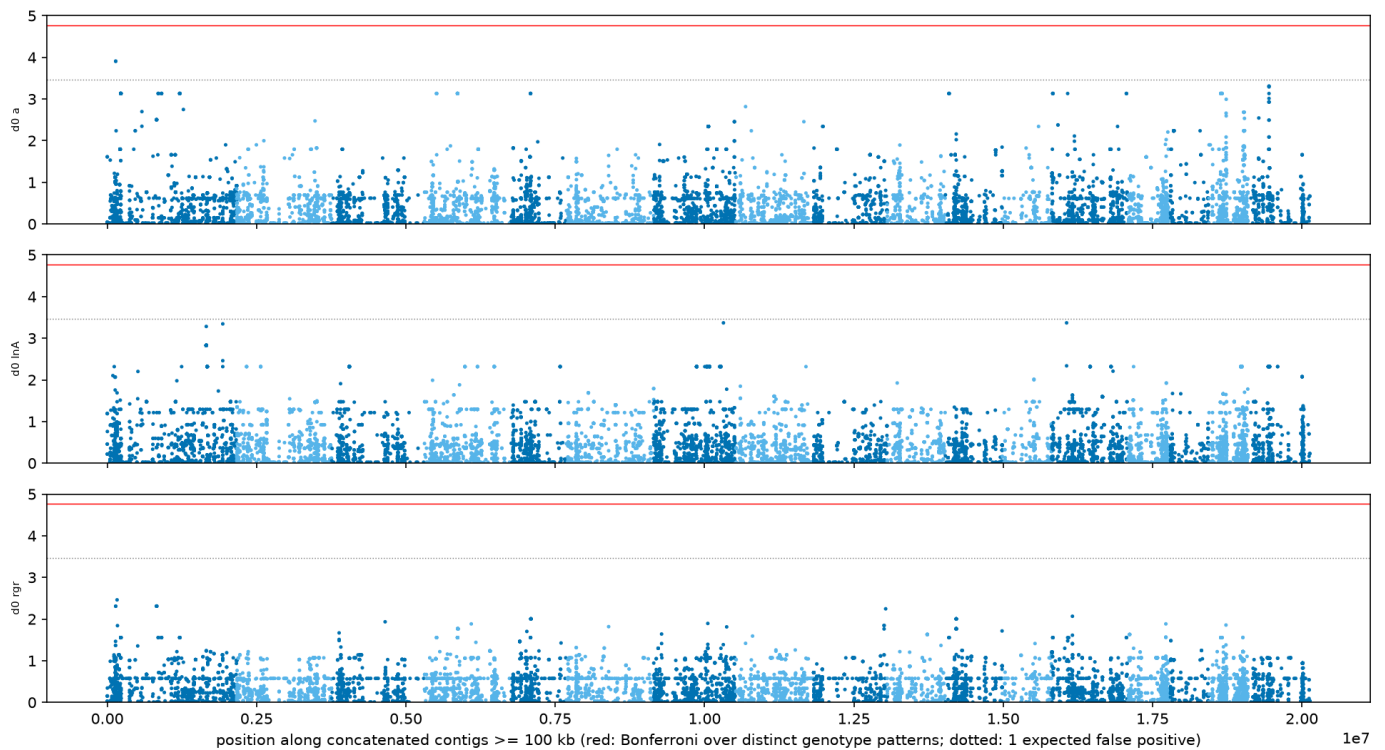


Figure 17. Dose-0 traits, run-adjusted, lineage covariates.

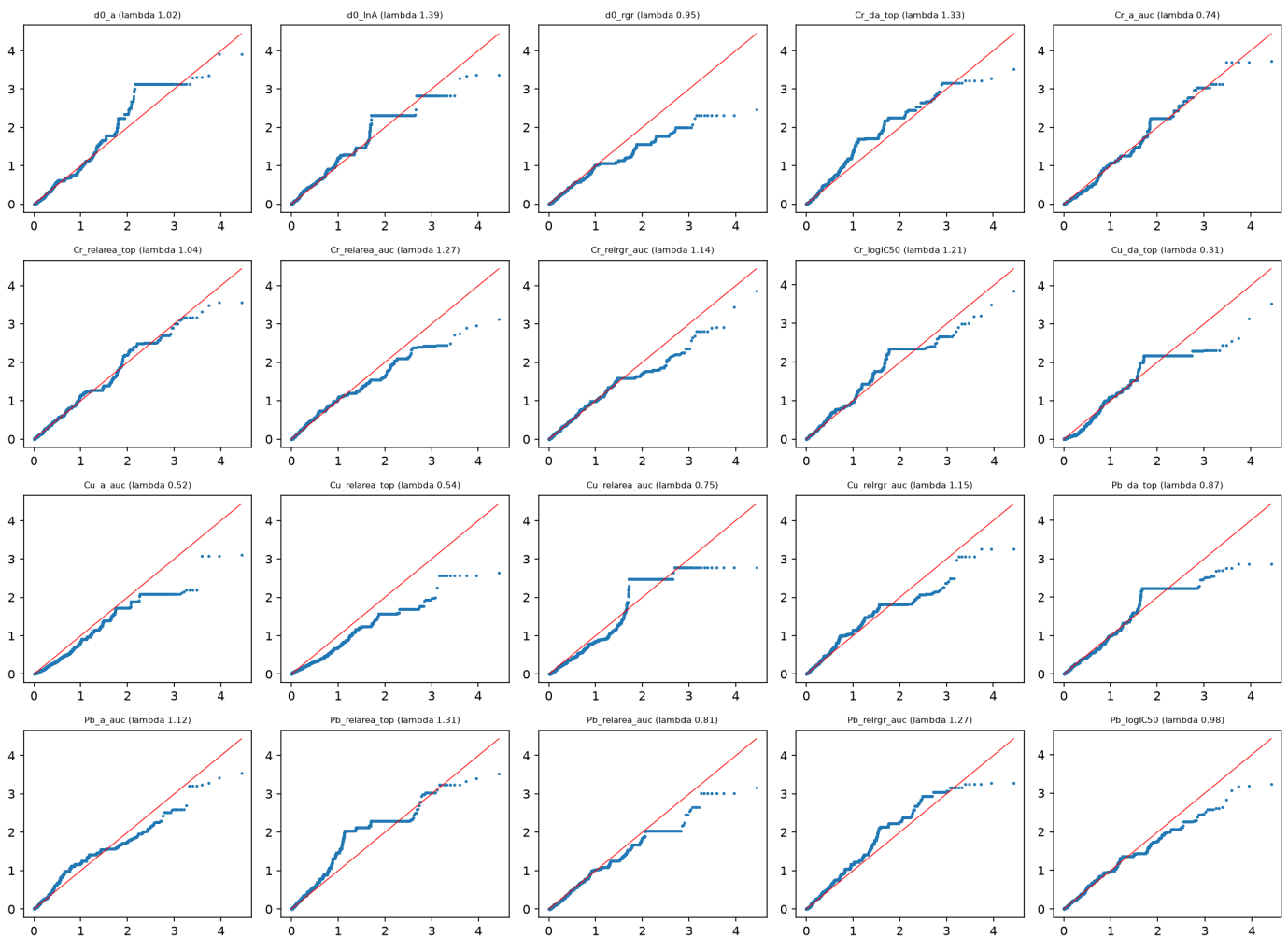


Figure 18. QQ plots of scan (c).

trait	chr	pos	af	beta	p_wald	effect	gene_id	product	distance_bp
d0_a	CM179485.1	141640	0.264	2.38	1.2e-04	upstream_gene_variant	ACY3AU_004087	cis-prenyltransferase	0
Cr_relgrain	CM179490.1	904292	0.053	-	1.4e-04	3_prime_UTR_variant	ACY3AU_005509	metical protein	0
Cr_logIC50	CM179486.1	55694	0.161	4.21	1.5e-04	upstream_gene_variant	ACY3AU_000323	metical protein	93
Cr_a_auc	CM179487.1	957384	0.168	-	1.9e-04	missense_variant	ACY3AU_005580	Nuclear control of ATPase 2	0
Cr_a_auc	CM179497.1	588055	0.211	-	2.0e-04	splice_region_variant	ACY3AU_003010	a mannosyltransferase of the dolichol pathway	0
Cr_relarea	CM179491.1	380349	0.06	-	2.7e-04	missense_variant	ACY3AU_000387	metical protein	0
Pb_a_auc	CM179495.1	344369	0.121	-	2.9e-04	missense_variant	ACY3AU_002286	metical protein	0
Cu_da_top	CM179485.1	1274990	0.132	1.63	2.9e-04	upstream_gene_variant	ACY3AU_004469	metical protein	20
Pb_relarea	CM179495.1	346567	0.059	-	2.9e-04	missense_variant	ACY3AU_002287	metical protein	0
Cr_da_top	CM179497.1	299036	0.127	1.92	3.0e-04	upstream_gene_variant	ACY3AU_000014	metical protein	2354
Cr_logIC50	JBZGVR010000430	500143	0.112	-	3.2e-04	upstream_gene_variant	ACY3AU_000808	metical protein	102
Cr_relarea	CM179489.1	1407586	0.053	-	3.3e-04	upstream_gene_variant	ACY3AU_000143	metical protein	410

Table 13. Suggestive loci of scan (c) (p below $1 / 2,940$). None passes the Bonferroni threshold ($p < 1.7e-5$). *effect* is the *snpEff* effect of the lead SNP; the *gene* is the one overlapping or nearest the lead SNP, with distance in bp.

- None of the 12 loci passes the threshold, and 17 SNPs reach the suggestive level across 20 traits, against about 94 expected by chance. The scan shows no excess of signal.
- Two loci lie 2 kb apart on contig CM179495.1 (Pb a* AUC, ACY3AU_002286, and Pb relative area, ACY3AU_002287, both missense variants; $p = 2.9e-4$ each). The two traits are different summaries of the same Pb data, and the result is not replicated in an independent experiment.
- Candidates with a named product are: ACY3AU_005580 (Nuclear control of ATPase 2, Cr a* AUC, missense), ACY3AU_003010 (a mannosyltransferase of the dolichol pathway, Cr a* AUC, splice-region variant) and ACY3AU_004087 (cis-prenyltransferase, dose-0 a*). Prenyl and dolichol pathway genes are of interest for a* biology, but this is a hypothesis for later work.
- The SNP heritability estimates that GEMMA reports with lineage covariates are at the boundary (0 or 1) and are not used.

6. Non-additive architecture

Strains are haploid, so there is no dominance. Non-additive variance here means epistasis. Two analyses:

- **Variance components (REML).** Model: additive kernel (centred relatedness) plus additive-by-additive kernel (its element-wise square) plus error. The epistatic component is tested by a likelihood ratio test with the boundary correction.
- **Pairwise scan.** All pairs of the 1,401 LD-pruned SNPs ($r^2 < 0.5$), interaction term in an LMM that is adjusted for relatedness. Pairs need at least 5 strains in each of the four two-locus classes. Between 51,888 and 65,081 of 980,700 pairs per trait meet this (the others are collinear or rare because of the lineage structure).

trait	n	h2_additive_only	h2_additive	h2_epistatic_AA	residual_share	LRT	p_epistatic
d0_a	125	0.264	0.000	0.216	0.784	1.711	0.095

trait	n	h2_additive_only	h2_additive	h2_epistatic_AA	residual_share	LRT	p_epistatic
d0_lnA	125	0.000	0.000	0.160	0.840	0.000	1.000
d0_rgr	125	0.034	0.034	0.000	0.966	0.000	0.500
Cr_da_top	125	0.091	0.056	0.037	0.907	0.191	0.331
Cr_a_auc	125	0.184	0.177	0.007	0.816	0.006	0.470
Cr_relarea_top	125	0.155	0.155	0.000	0.845	0.000	0.500
Cr_relarea_auc	125	0.129	0.123	0.006	0.871	0.003	0.479
Cr_relrgr_auc	124	0.058	0.000	0.062	0.938	0.893	0.172
Cr_logIC50	123	0.045	0.045	0.000	0.955	0.000	0.500
Cu_da_top	125	0.177	0.173	0.004	0.824	0.001	0.489
Cu_a_auc	125	0.228	0.228	0.000	0.772	0.000	0.500
Cu_relarea_top	125	0.162	0.162	0.000	0.838	0.000	0.500
Cu_relarea_auc	125	0.065	0.027	0.043	0.930	0.149	0.350
Cu_relrgr_auc	125	0.000	0.000	0.012	0.988	0.311	0.289
Pb_da_top	123	0.087	0.012	0.144	0.844	0.963	0.163
Pb_a_auc	123	0.043	0.000	0.051	0.949	1.243	0.132
Pb_relarea_top	123	0.000	0.000	0.000	1.000	0.000	1.000
Pb_relarea_auc	123	0.006	0.006	0.000	0.994	0.000	0.500
Pb_relrgr_auc	122	0.000	0.000	0.000	1.000	0.000	1.000
Pb_logIC50	119	0.060	0.060	0.000	0.940	0.000	0.500

Table 14. Variance shares from the REML models (RINT traits, panel strains). **h2_additive_only** is the SNP heritability with the additive kernel alone.

trait	n_pairs_tested	bonferroni_threshold	min_p	lambda_gc	n_bonferroni	n_expected_at_1e_5	n_at_1e_5
d0_a	65081	7.68e-07	3.1e-04	0.909	0	0.651	0
d0_lnA	65081	7.68e-07	1.3e-04	0.986	0	0.651	0
d0_rgr	65081	7.68e-07	4.0e-04	0.63	0	0.651	0
Cr_da_top	65081	7.68e-07	0.001	0.952	0	0.651	0
Cr_a_auc	65081	7.68e-07	0.002	1.17	0	0.651	0
Cr_relarea_top	65081	7.68e-07	1.9e-04	0.88	0	0.651	0
Cr_relarea_auc	65081	7.68e-07	7.4e-05	1.17	0	0.651	0
Cr_relrgr_auc	65081	7.68e-07	8.5e-04	1.73	0	0.651	0
Cr_logIC50	62079	8.05e-07	7.3e-05	1.11	0	0.621	0
Cu_da_top	65081	7.68e-07	6.1e-04	1.06	0	0.651	0
Cu_a_auc	65081	7.68e-07	1.1e-04	0.755	0	0.651	0
Cu_relarea_top	65081	7.68e-07	8.2e-04	0.525	0	0.651	0
Cu_relarea_auc	65081	7.68e-07	4.9e-05	1.04	0	0.651	0
Cu_relrgr_auc	65081	7.68e-07	6.6e-04	1.56	0	0.651	0
Pb_da_top	64402	7.76e-07	2.3e-04	1.1	0	0.644	0
Pb_a_auc	64402	7.76e-07	1.5e-04	1.21	0	0.644	0
Pb_relarea_top	64402	7.76e-07	0.003	1.46	0	0.644	0

trait	n_pairs_tested	bonferroni_threshold	min_p	lambda_gc	n_bonferroni	n_expected_at_1e_5	n_at_1e_5
Pb_relarea_auc	64402	7.76e-07	0.002	0.768	0	0.644	0
Pb_relgr_auc	64402	7.76e-07	0.002	0.596	0	0.644	0
Pb_logIC50	51888	9.64e-07	8.0e-05	0.99	0	0.519	0

Table 15. Pairwise interaction scan. No pair passes the Bonferroni threshold. **n_expected_at_1e_5** is the number expected below 1e-5 by chance.

- SNP heritability from the additive kernel alone is 0 to 0.26: d0 a* 0.26, Cu a* AUC 0.23, Cr a* AUC 0.18, Cu a* change 0.18, Cr and Cu relative area at the top dose 0.16. Standard errors are 0.08 to 0.17 (GEMMA, `gwas_inflation_pve_unadjusted.csv`).
- The epistatic component is zero or small in all traits. The smallest p is 0.095 (d0 a*). When it is non-zero it takes variance away from the additive component (d0 a*: 0.26 additive only, 0.00 additive and 0.22 epistatic). The two kernels are highly correlated in a clonal panel, so the split between them is not identifiable.
- The pairwise scan has no hit and no excess (lambda 0.5 to 1.7; median 1.02).
- With 14 lineages, no analysis can tell an additive lineage effect from an epistatic one. Both are a difference between whole genotypes.

7. Recombination inside the large lineages

Informative SNPs are those with minor allele count of at least 2 inside the lineage (all SNPs of the QC VCF, not only common ones): 1,691 in L01 (42 strains), 2,756 in L02 (34) and 1,661 in L03 (16). Two tests:

- **Four-gamete test and r^2 by distance.** Under strict clonality every pair of sites is equally linked, so the fraction of pairs with all four two-locus classes does not depend on distance. Under recombination close pairs stay compatible and unlinked pairs (other contigs) become incompatible.
- **Windowed distance matrices.** For 300 kb windows the strain-by-strain SNP distance matrix is compared between windows. With recombination, distant windows would show different relationships between strains than adjacent windows.

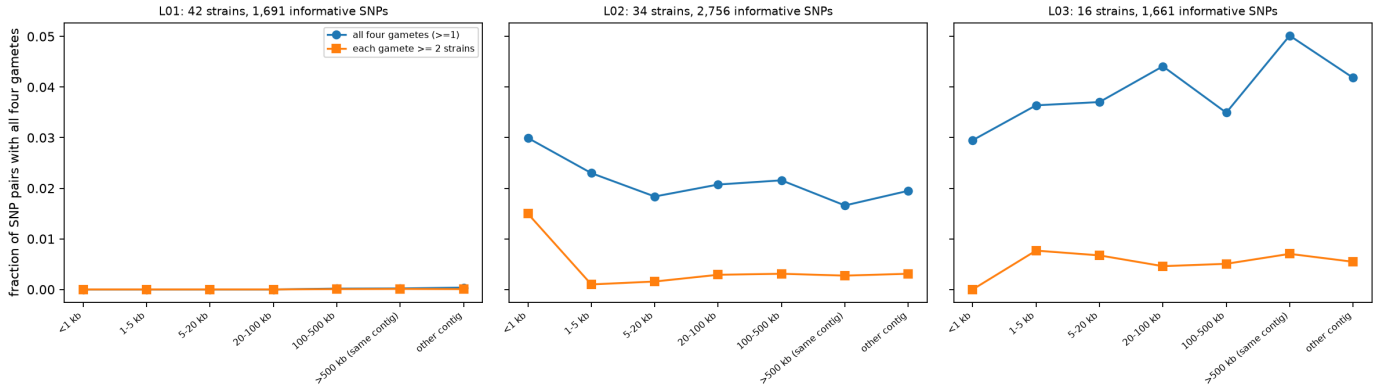


Figure 19. Fraction of SNP pairs with all four gametes, by physical distance (and pairs on different contigs), for the three largest lineages.

lineage	n_strains	n_snps	pair_class	n_pairs	mean_r2	frac_four_gametes	frac_four_gametes_each_ge2
L01	42	1691	<1 kb	3025	0.265	0.000	0.000
L01	42	1691	1-5 kb	8099	0.166	0.000	0.000
L01	42	1691	5-20 kb	29676	0.144	0.000	0.000
L01	42	1691	20-100 kb	30000	0.150	0.000	0.000
L01	42	1691	100-500 kb	30000	0.151	0.000	0.000
L01	42	1691	>500 kb (same contig)	30000	0.147	0.000	0.000
L01	42	1691	other contig	30000	0.153	0.000	0.000
L02	34	2756	<1 kb	3407	0.290	0.030	0.015

lineage	n_strains	n_snps	pair_class	n_pairs	mean_r2	frac_four_gametes	frac_four_gametes_each_ge2
L02	34	2756	1-5 kb	9047	0.130	0.023	0.001
L02	34	2756	5-20 kb	30000	0.116	0.018	0.002
L02	34	2756	20-100 kb	30000	0.122	0.021	0.003
L02	34	2756	100-500 kb	30000	0.120	0.022	0.003
L02	34	2756	>500 kb (same contig)	30000	0.117	0.017	0.003
L02	34	2756	other contig	30000	0.120	0.019	0.003
L03	16	1661	<1 kb	3696	0.768	0.029	0.000
L03	16	1661	1-5 kb	9509	0.550	0.036	0.008
L03	16	1661	5-20 kb	30000	0.544	0.037	0.007
L03	16	1661	20-100 kb	30000	0.563	0.044	0.005
L03	16	1661	100-500 kb	30000	0.576	0.035	0.005
L03	16	1661	>500 kb (same contig)	30000	0.540	0.050	0.007
L03	16	1661	other contig	30000	0.557	0.042	0.005

Table 16. Pair classes: mean r^2 , fraction of pairs with all four gametes (each gamete in at least 1 strain, and in at least 2 strains).

lineage	window_pair_class	n_window_pairs	mean_spearman	q05	n_windows
L01	adjacent windows (same contig)	83	0.64	0.39	64
L01	distant windows (same contig, > 1 Mb at 300 kb windows)	20	0.58	0.29	64
L01	different contigs	3800	0.68	0.41	64
L02	adjacent windows (same contig)	92	0.58	0.21	66
L02	distant windows (same contig, > 1 Mb at 300 kb windows)	26	0.59	0.32	66
L02	different contigs	3777	0.60	0.35	66
L03	adjacent windows (same contig)	92	0.83	0.60	63
L03	distant windows (same contig, > 1 Mb at 300 kb windows)	18	0.86	0.74	63
L03	different contigs	3795	0.81	0.60	63

Table 17. Spearman correlation of strain-by-strain distance matrices between 300 kb windows.

Reading.

- **L01 has no incompatible pair at any distance** (0 of about 190,000 sampled pairs). Its r^2 is 0.27 below 1 kb and 0.14 to 0.15 at every other distance, including other contigs.
- **L02 and L03 have a few percent incompatible pairs, with no trend with distance** (L02: 3.0% below 1 kb, 1.7% to 2.3% beyond; 2.0% on other contigs). Requiring each gamete in at least 2 strains lowers this to 0.1% to 0.3% beyond 1 kb in L02 (1.5% below 1 kb) and 0.5% to 0.8% in L03 (0% below 1 kb). The rate on other contigs equals the rate within contigs. This is the signature of recurrent mutation or genotype errors, not of recombination.
- **The distance-matrix correlation does not depend on window distance** (L01 0.64 adjacent, 0.58 distant, 0.68 other contigs; L02 0.58, 0.59, 0.60; L03 0.83, 0.86, 0.81). The strains have the same relationships at every position.
- **A small excess of r^2 below 1 kb** (0.27 in L01, 0.29 in L02) is visible. This may come from clustered mutations or mapping errors. It does not extend beyond 1 kb.
- **Conclusion.** There is no evidence of recombination in these lineages. They behave as clones that accumulated mutations. Rare events (for example gene conversion tracts shorter than 1 kb or a few recombinants among many strains) cannot be excluded with these tests.

8. The Cr tree signal

Metal	trait	n_strains	n_runs	pagel_lambda	p_lambda_gt_0
Chromium	baseline a*	163	4	0.912	0.008
Chromium	baseline a*, run effect removed	163	4	0.886	0.117
Copper	baseline a*	163	4	0.000	1.000
Copper	baseline a*, run effect removed	163	4	0.000	1.000
Lead	baseline a*	161	4	0.000	1.000
Lead	baseline a*, run effect removed	161	4	0.000	1.000

Table 18. *Pagel's lambda of baseline a* in pure haploid R. mucilaginosa** (outgroup-rooted PHYling tree), before and after removing run effects (run mean subtracted).*

Metal	factor	n_groups	n_strains	r2	adj_r2
Chromium	run	4	163	0.156	0.140
Chromium	lineage	14	122	0.242	0.150
Copper	run	4	163	0.037	0.019
Copper	lineage	14	122	0.220	0.127
Lead	run	4	161	0.031	0.013
Lead	lineage	14	120	0.215	0.119

Table 19. *Share of variance in strain-mean baseline a* explained by run and by lineage (adjusted R²). Lineage needs a lineage label, so n differs.*

- **Cr:** lambda is 0.91 ($p = 0.008$, $n = 163$). After removing run means it is 0.89 ($p = 0.12$). The point estimate hardly changes, but the evidence for lambda above 0 weakens, so the signal is not robust. **Cu and Pb:** lambda is 0 in both versions.
- **Run matters for Cr.** Mean baseline a* by run: d000320 20.9, d000321 21.2, d000322 19.6 (SD 3.9), d000323 18.1. Run explains 14% of the variance in strain means (Cu 2%, Pb 1%). Lineage explains 15% (Cu 13%, Pb 12%).
- **Low-a* lineages sit in late runs.** In Cr, L05 (mean a* 15.6) has 5 of 6 strains in run d000322, and L07 (16.0) has 6 of 6 in d000322 or d000323. The large lineages L01 (20.1) and L02 (20.7) are spread over all four runs. So a few lineages with low a* coincide with the runs with low a*. Two explanations fit: a genetic difference of those lineages, or a run effect. The data cannot separate them.
- **Cr lineage structure on the tree.** The tree signal comes mostly from the separation between a small number of lineages and clades, consistent with section 4. It does not show a continuous relationship between relatedness and trait inside the species.

9. Candidate SNP and gene panels, with window context (Cr)

Panel A: lineage-level partitions. The Cr Bonferroni loci of the unadjusted scan (the run-adjusted scan has none for Cr) fall into a few genotype partitions, each carried by a set of whole lineages. For each partition the table counts the SNPs that share its genotype pattern and the genes with high or moderate snpEff effects among them.

partition	lead_trait	lead_snp	strains_with_lead	lineages_with_lead	n_perfectly_linked_high	n_perfectly_linked_moderate	genes_high	genes_moderate
P1	Cr_relarea_auc [unadjusted]	JBZGVR0100000152	22:179652	G1A05,L06,L07,L09,L117,L12,L14	0	0	0	
P2	Cr_relarea_auc [unadjusted]	CM179490.1:9816205	C:T	L04,L05,L06,L07,L095,L12,L14	1	7	1	
P3	Cr_relarea_auc [unadjusted]	CM179490.1:1242638	A:G	L04,L05,L06,L07,L081,B09,L12,L14	0	20	3	
P4	Cr_relarea_auc [unadjusted]	CM179492.1:9818129	T:G	L04,L05,L06,L07,L097,L10,L12,L14	0	10	3	
P5	Cr_relarea_auc [unadjusted]	CM179490.1:9093840	C:T	L04,L05,L06,L07,L091,L12,L13,L14	0	0	0	
P6	Cr_relarea_auc [unadjusted]	CM179490.1:1320806	A:G	L05,L07,L09,L12,L14124	0	24	6	

partition	lead_trait	lead_snp	strains_within_lineages	genes_with_alt	n_perfectly_linked_snps	high	moderate	genes_high_moderate
P7	Cr_relarea_auc [unadjusted]	CM179486.1:1276922A:C	L04,L05,L07,L09,L12,L14	14	0	20	6	
P8	Cr_relarea_top [unadjusted]	CM179496.1:3019730A	L02,L05,L06,L07,L09,L10,L13,L14	14	3	92	12	

Table 20. Cr lineage partitions (lead trait and scan in brackets). **n_perfectly_linked_snps** counts SNPs with the same genotype pattern.

Panel B: within-lineage candidates. SNPs with $p < 1e-3$ in scan (c) for the six Cr traits, and SNPs with $p < 0.01$ in the per-lineage permutation scans (L01, L02, L03). The full table is `report/tables/cr_candidate_snp_panel.csv` (91 rows, 74 SNPs, 175 genes within 5 kb).

source	trait	chr	pos	p	p_family_effect	genes_within_5kb	products
scan c (all lineages,	Cr_relrgrCM179490.1042921.4e-04				3_prime_UTR_variant	ACY3AU_001506; ACY3AU_001507; ACY3AU_001	hypothetical protein; hypothetical protein; hypothetical protein; hypo
scan c (all lineages,	Cr_logIC50CM179486.1556941.5e-04				upstream_gene_variant	ACY3AU_000320; ACY3AU_000321; ACY3AU_000	hypothetical protein; hypothetical protein; hypo
scan c (all lineages,	Cr_a_aucCM179487.19573841.9e-04				missense_variant	ACY3AU_005578; ACY3AU_005579; ACY3AU_005	hypothetical protein; white collar 2 type of transcription fac; Nuclea
scan c (all lineages,	Cr_a_aucCM179497.15880562.0e-04				splice_region_variant	ACY3AU_003009; ACY3AU_003010; ACY3AU_003	carnitine O-acetyltransferase yat1; dolichyl-P-Man:Man(5)GlcNAc(2)-PP-
scan c (all lineages,	Cr_a_aucCM179497.15880562.0e-04				splice_region_variant	ACY3AU_003009; ACY3AU_003010; ACY3AU_003	carnitine O-acetyltransferase yat1; dolichyl-P-Man:Man(5)GlcNAc(2)-PP-
scan c (all lineages,	Cr_a_aucCM179497.15880832.0e-04				upstream_gene_variant	ACY3AU_003009; ACY3AU_003010; ACY3AU_003	carnitine O-acetyltransferase yat1; dolichyl-P-Man:Man(5)GlcNAc(2)-PP-
scan c (all lineages,	Cr_a_aucCM179497.15880562.0e-04				splice_region_variant	ACY3AU_003009; ACY3AU_003010; ACY3AU_003	carnitine O-acetyltransferase yat1; dolichyl-P-Man:Man(5)GlcNAc(2)-PP-
scan c (all lineages,	Cr_relareaCM179491.1803492.7e-04				missense_variant	ACY3AU_004885; ACY3AU_004886; ACY3AU_004	WD domain protein; Peptidyl-prolyl cis-trans isomerase B, v; hypotheti

source	trait	chr	pos	p	p_family_wise	effect	genes_within_5kb	products
scan c (all lineages,	Cr_relarea_top	CM179491.1	380352.7e-04			synonymous_variant	ACY3AU_004885; ACY3AU_004886; ACY3AU_004	WD domain protein; Peptidyl-prolyl cis-trans isomerase B, v; hypotheti
scan c (all lineages,	Cr_da_top	CM179497.2	299036.0e-04			upstream_gene_variant	ACY3AU_002913; ACY3AU_002914; ACY3AU_002	Nucleolar protein 16; hypothetical protein; hypothetical protein
scan c (all lineages,	Cr_logIC50	BZGVR0150030352e				upstream_gene_variant	ACY3AU_002806; ACY3AU_002807; ACY3AU_002	hypothetical protein; hypothetical protein; hypothetical protein; hypo
scan c (all lineages,	Cr_relarea_top	CM179489.1	407586.3e-04			upstream_gene_variant	ACY3AU_002141; ACY3AU_002142; ACY3AU_002	hypothetical protein; hypothetical protein; hypothetical protein; hypo
scan c (all lineages,	Cr_relgr_Cmt	CM179486.1	886 3.6e-04			5_prime_UTR_variant	ACY3AU_000305; ACY3AU_000306; ACY3AU_000	RNA polymerase III subunit C82; Transmembrane nucleoporin; hypothetica
within L02 (permutatio	Cr_relarea_top	CM179495.1	20604.6e-04	0.009		upstream_gene_variant	ACY3AU_002196; ACY3AU_002197; ACY3AU_002	Sister chromatid cohesion protein 2; Actin-related protein 4; hypothet
within L02 (permutatio	Cr_relarea_top	CM179495.1	20654.6e-04	0.009		upstream_gene_variant	ACY3AU_002196; ACY3AU_002197; ACY3AU_002	Sister chromatid cohesion protein 2; Actin-related protein 4; hypothet

Table 21. The 15 strongest rows of panel B. Genes within 5 kb of the SNP are listed with their products.

lineage	trait	n_strains	n_snps	distinct_patterns	min_p	min_p_family_wise
L01	Cr_da_top	42	1691	41	0.035	0.629
L01	Cr_a_auc	42	1691	41	0.020	0.42
L01	Cr_relarea_top	42	1691	41	0.078	0.911
L01	Cr_relarea_auc	42	1691	41	0.013	0.274
L01	Cr_relrgr_auc	41	1691	41	0.004	0.233
L01	Cr_logIC50	42	1691	41	0.063	0.861
L02	Cr_da_top	33	2756	121	0.006	0.379
L02	Cr_a_auc	33	2756	121	0.002	0.136
L02	Cr_relarea_top	33	2756	121	4.6e-04	0.0085
L02	Cr_relarea_auc	33	2756	121	0.008	0.362
L02	Cr_relrgr_auc	33	2756	121	0.019	0.747
L02	Cr_logIC50	33	2756	121	0.013	0.68
L03	Cr_da_top	16	1661	110	0.002	0.221
L03	Cr_a_auc	16	1661	110	0.004	0.38
L03	Cr_relarea_top	16	1661	110	0.012	0.57
L03	Cr_relarea_auc	16	1661	110	0.011	0.543
L03	Cr_relrgr_auc	16	1661	110	0.013	0.695
L03	Cr_logIC50	16	1661	110	0.007	0.325

Table 22. Per-lineage permutation scans for *Cr* traits (family-wise *p* from 2,000 permutations of the trait across strains).

genes_with_snps	keyword_genes	mean_neglog10p_keyword	mean_neglog10p_all	perm_p
3605	107	1.17	1.21	0.777

Table 23. Are genes with a metal, stress or pigment-related product keyword enriched among the *Cr* scan (c) signal? Mean $-\log_{10} p$ of the best SNP in or within 2 kb of each gene, keyword genes against all genes with SNPs; *p* from 10,000 random gene sets of the same size.

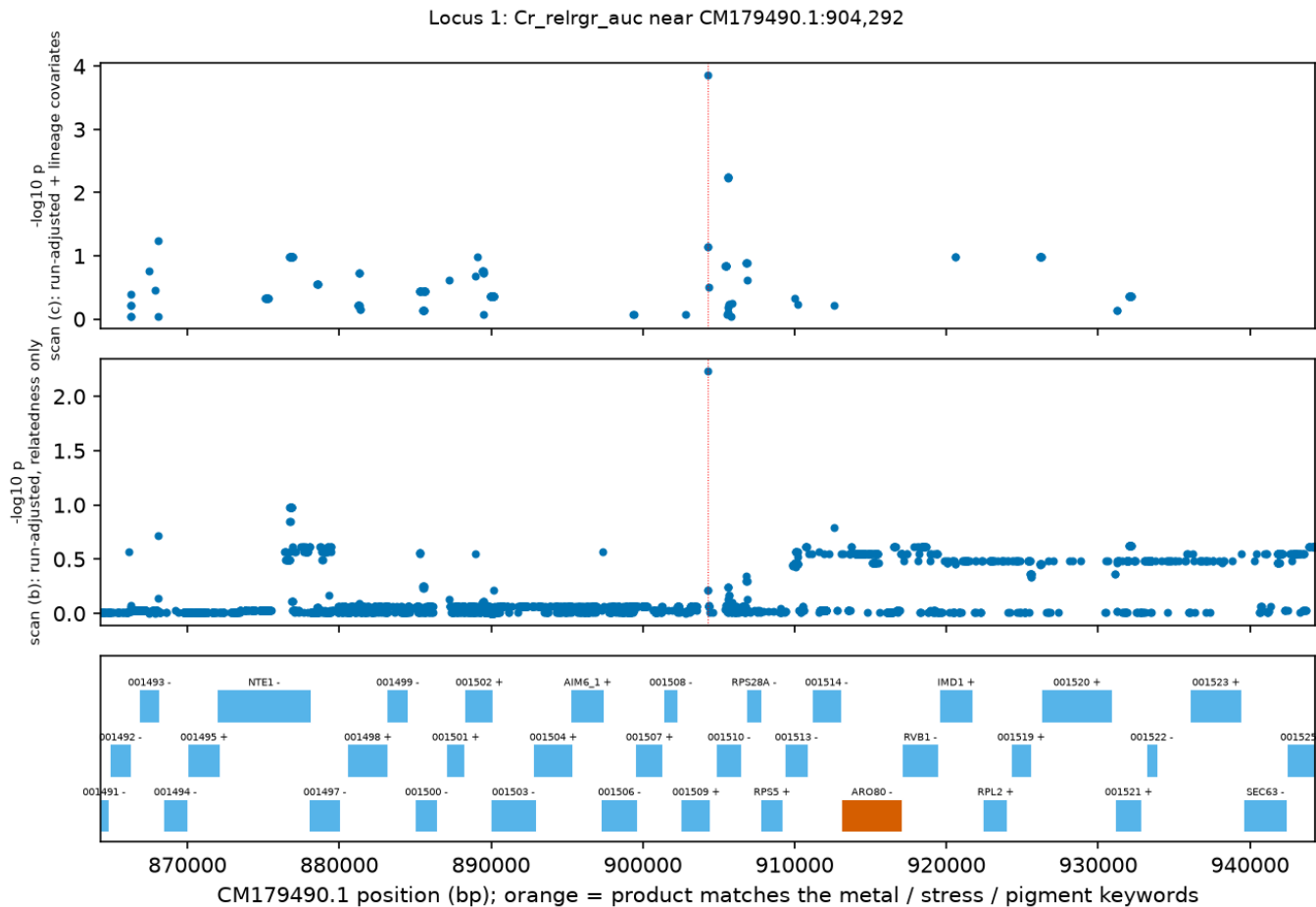


Figure 20. Locus 1: *Cr* relative growth-rate AUC near CM179490.1:904,292. Top: scan (c). Middle: scan (b). Bottom: gene models in the window (orange: product matches the keyword list).

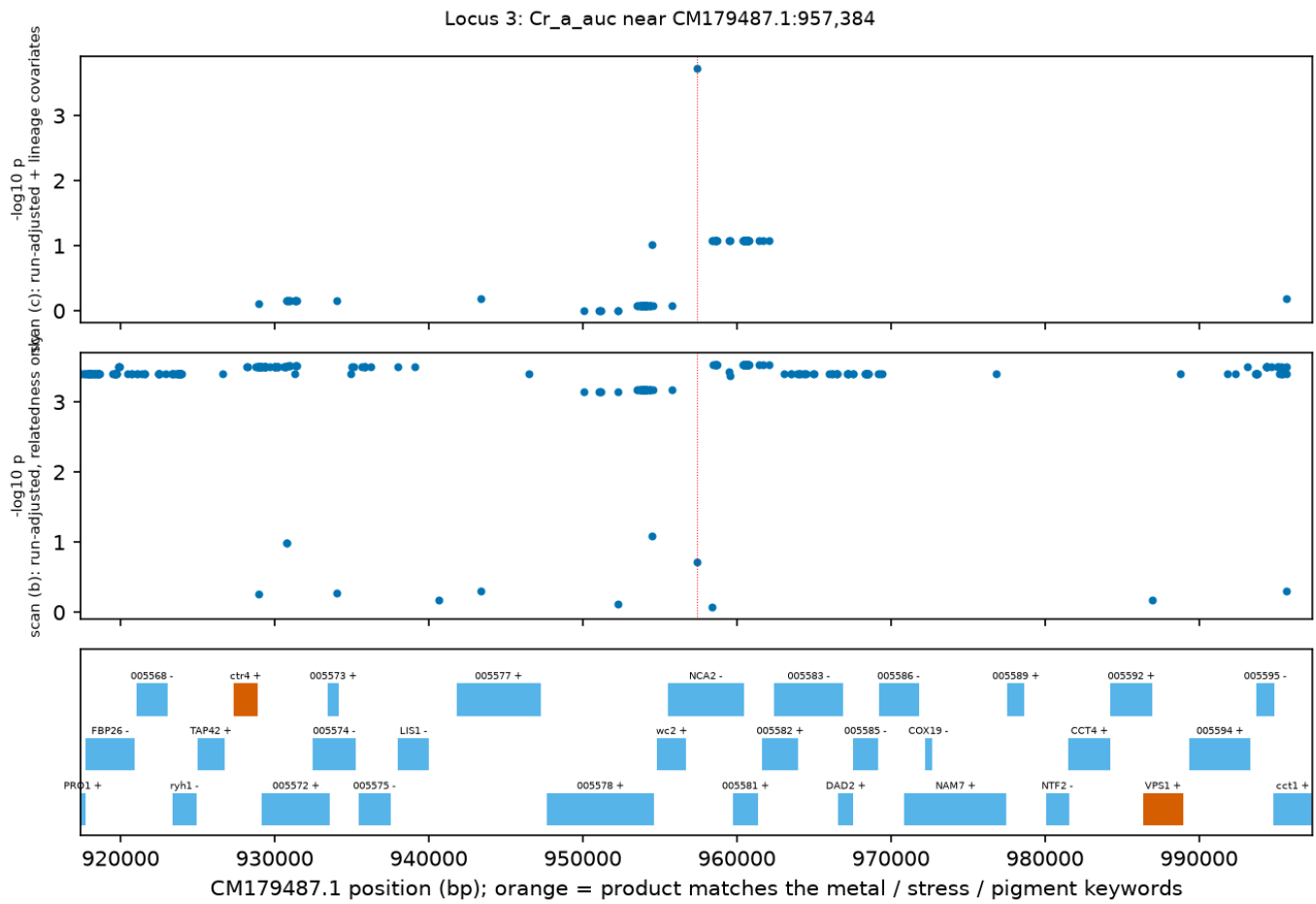


Figure 21. Locus 3: Cr a | AUC near CM179487.1:957,384 (missense variant in ACY3AU_005580, Nuclear control of ATPase 2).*

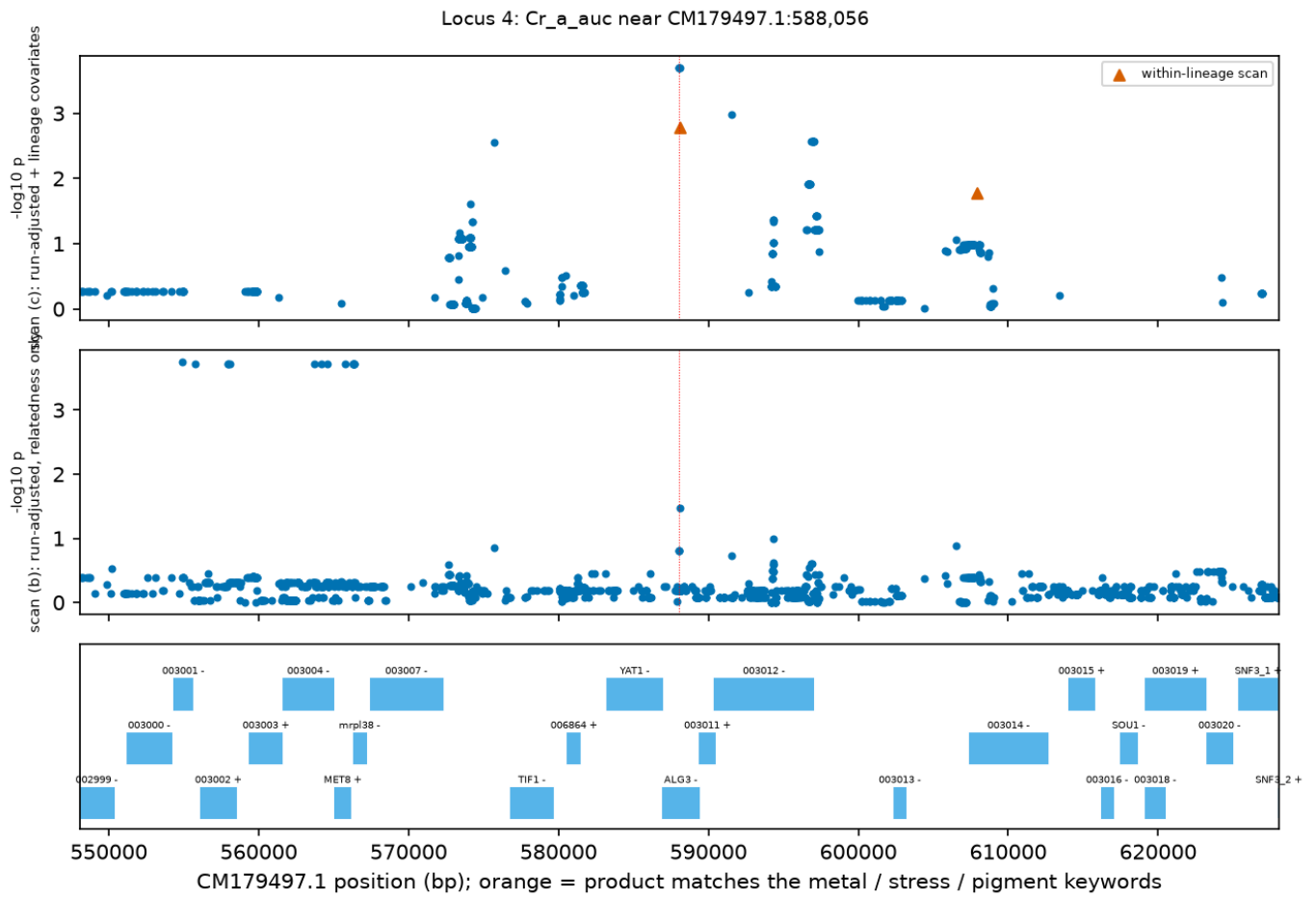


Figure 22. Locus 4: Cr a | AUC near CM179497.1:588,056 (splice-region variants in ACY3AU_003010).*

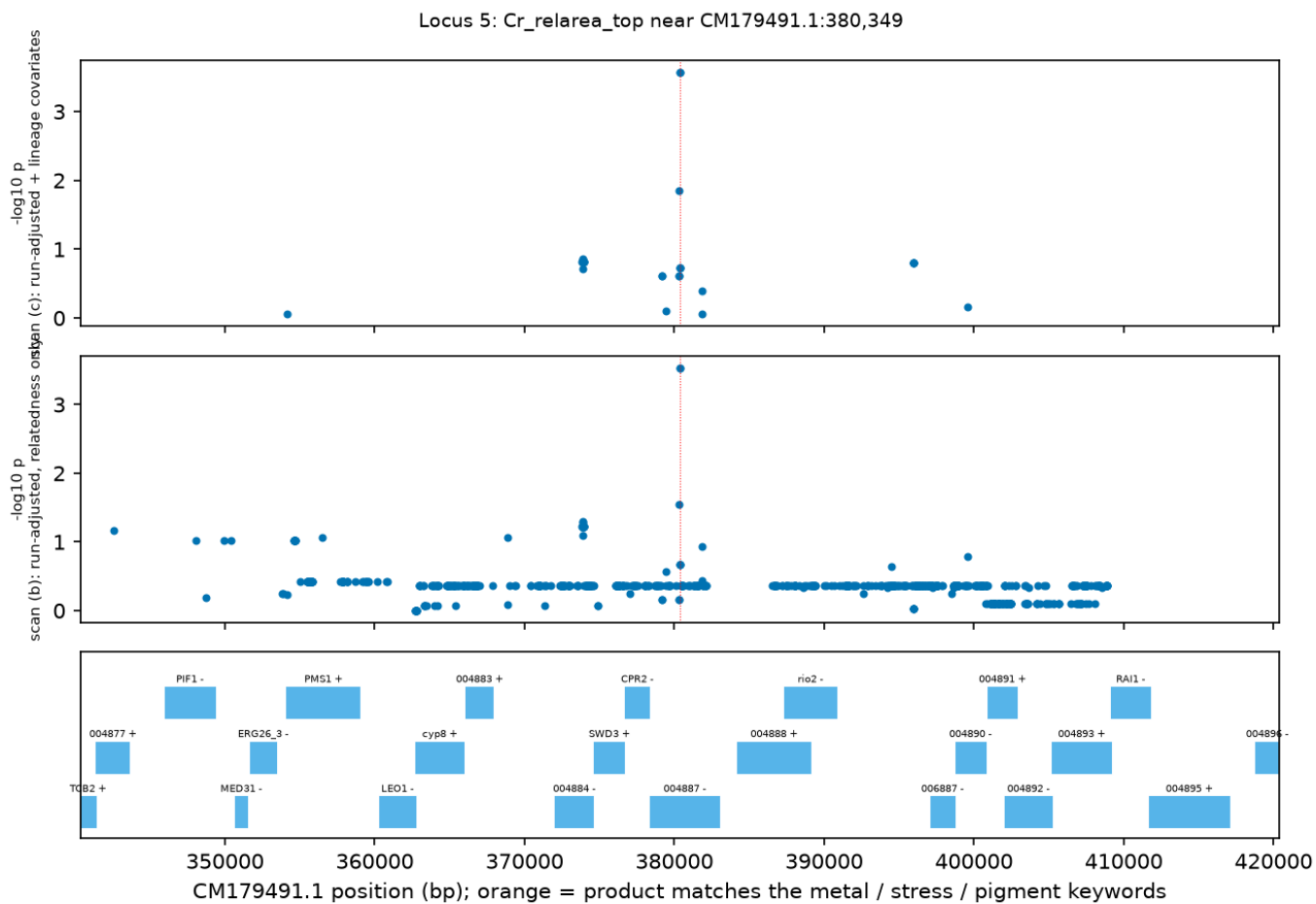


Figure 23. Locus 5: Cr relative area at the top dose near CM179491.1:380,349.

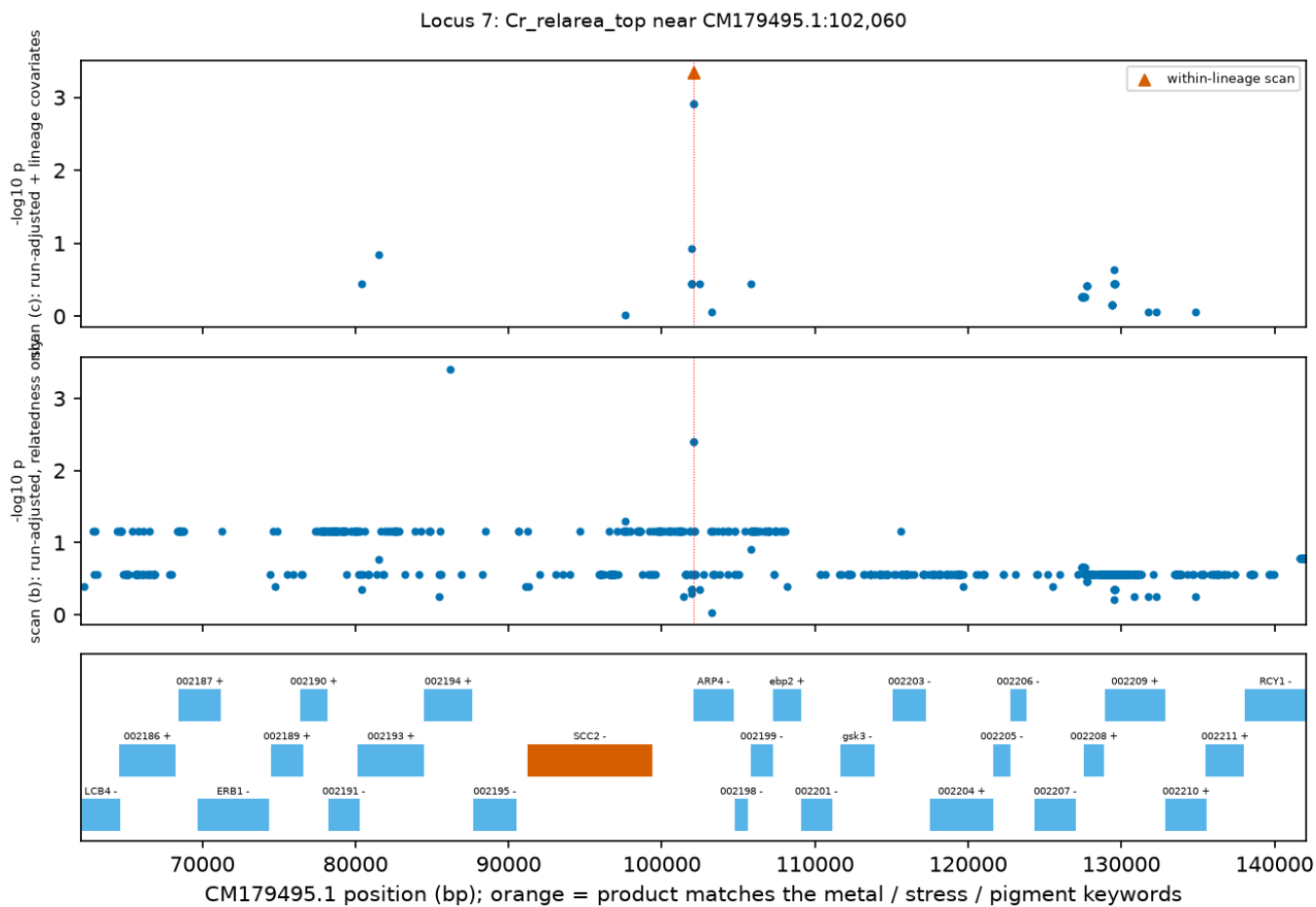


Figure 24. Locus 7: the only family-wise hit of the *Cr* within-lineage scans, in L02 (*Cr* relative area at the top dose, CM179495.1:102,060; 15 of 33 strains carry the alternate allele).

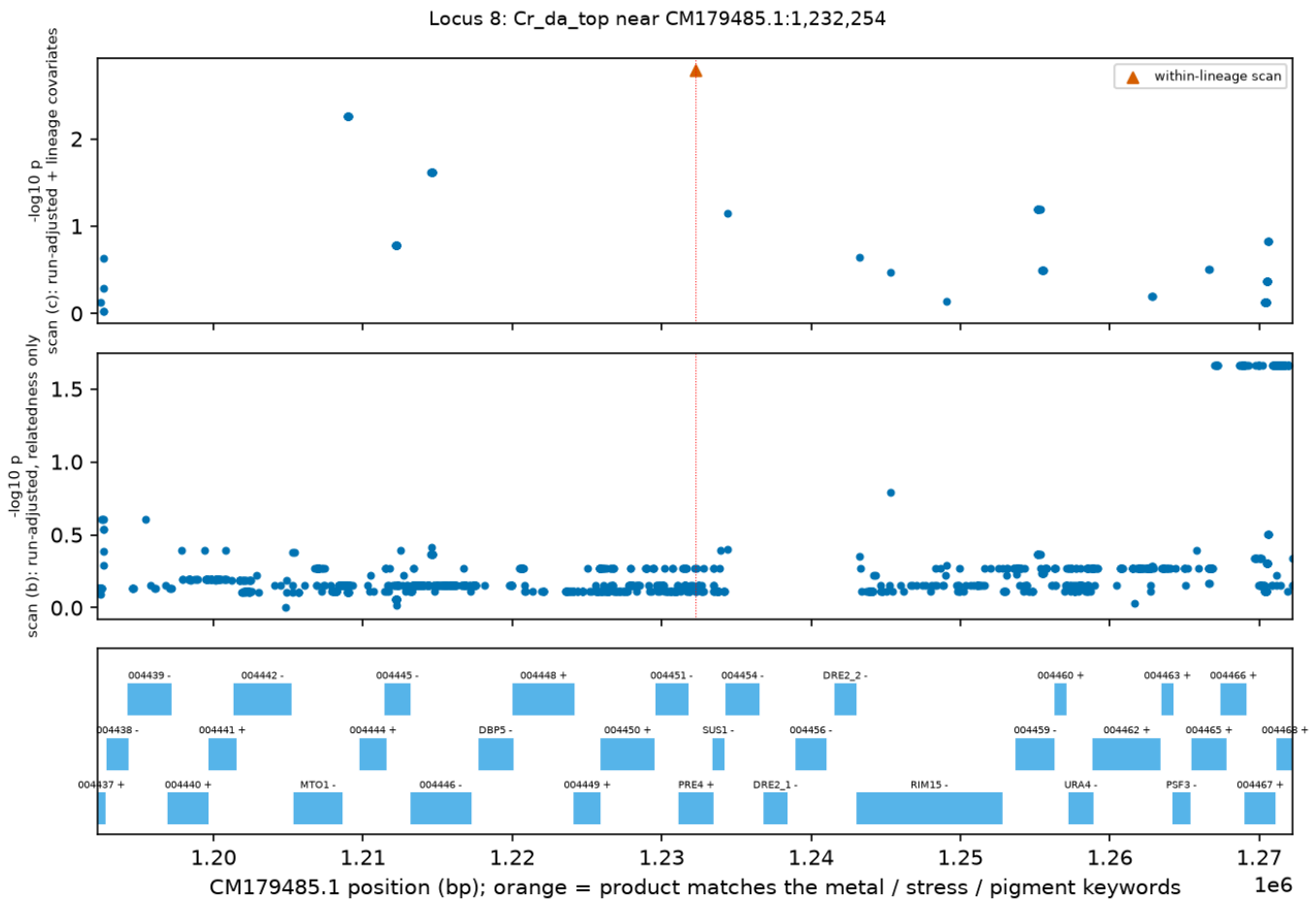


Figure 25. Locus 8: best Cr SNP of the L03 scan (16 strains).

Reading.

- **Lineage level.** The Cr traits differ among lineages (ICC 0.28 for relative area). The partitions that track the difference are carried by 16 to 50 strains from 5 to 8 lineages. Each has 7 to 619 perfectly linked SNPs and up to 12 genes with high or moderate effects (partition P8: 3 high, 92 moderate). The data cannot say which SNP matters, and cannot tell a genetic effect from a run effect (section 8).
- **Within lineages.** Only one Cr test reaches a family-wise p below 0.05: L02, relative area at the top dose, 2 SNPs 5 bp apart ($p = 4.6e-4$, family-wise 0.0085) upstream of ACY3AU_002196 (sister chromatid cohesion protein 2). Over the 60 lineage-by-trait tests, 3 have a family-wise p below 0.05 (the other two are Cu and Pb traits in L02), which equals the 3 expected by chance. With Bonferroni over 60 tests the best p is 0.51. L01 and L03 have none for Cr.
- **Window context.** In the zoom plots the SNPs at the top of each locus are single SNPs or a small cluster with no supporting neighbours. A true causal locus would show a peak of SNPs in linkage with the lead SNP. Here the lead SNPs are mostly isolated points, which is the pattern of noise.
- **Keyword genes.** 107 of 3,605 genes with SNPs have a metal, stress or pigment-related product. Their mean signal is not higher than for other genes (1.17 against 1.21; permutation $p = 0.78$).
- **Is there enough signal?** No, not in this panel: Cr traits clearly differ among lineages, but no locus can be assigned, and clone-corrected analysis inside Rmuc shows no result that survives correction.

10. Limits and next steps

- **Panels in section 9 are hypotheses.** None survives correction.
- **The panel cannot support gene mapping.** Every genome-wide signal tracks lineage. A usable panel needs many more independent lineages or recombinant strains (crosses, or populations with sexual recombination). D-49 (5-SNP cutoff) should be revisited: at 2,000 SNPs only 14 strains remain.
- **Run is confounded with lineage** (all screens). Run adjustment is therefore partial. A design with lineage spread over runs, or repeated runs, would separate them.

- **Replicate noise.** The strain share of the dose-0 variance is 38% for a* and lower for size and growth. Strain-level traits from 1 to 4 wells per dose are noisy, which lowers power further.
- **Cu IC50** is censored for 46% of panel strains; only coarse.
- **Fe and Zn** are held out (D-56). When complete tables exist, add them to METALS in 01_phenotypes.py.
- **Population analyses** (DH4148 SNP clusters, D-51) are not part of this report. The lineage table (results/lineages.csv) is a first version of that grouping.
- **Annotation.** Gene products come from the assembly GFF only. A GO and KEGG annotation of DH4148 would help to interpret candidates.
- **Scope of claims.** Candidate genes in section 5 are suggestive only. They have not been replicated.